



■ CS177 Bioinformatics: Software Development and Applications

Lecture 18: Other Omics

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Learning objectives



- At the end of this lecture, you should be able to:
 - Explain biological background of epigenomics
 - Describe ChIP-seq technique and related data analysis
 - Explain basics of ATAC-seq technique
 - Explain basics of Hi-C technique and data
 - Describe biological background of proteomics
 - Explain how mass spectrometry works
 - Define computational problems in proteomics
 - Define what is metabolomics
 - Describe the workflows of metabolomics studies





Biological Background of Epigenomics





Epigenetics



- **Epigenetics** is the science for studying **heritable** changes of gene expression, without involving changes of the **DNA sequence** itself
- There are at least two forms of information in the genome of the cell:
 - **Genetic information:** provides the building block for the manufacture of all proteins needed for the cell functional activity
 - **Epigenetic information:** provides additional instructions on how, when and where the genetic information should be used



Chromatin organization

- Multiple levels of packing are required to fit the DNA into the cell nucleus
- Net result: each DNA molecule has been packaged into a mitotic chromosome that is 10,000-fold shorter than its extended length

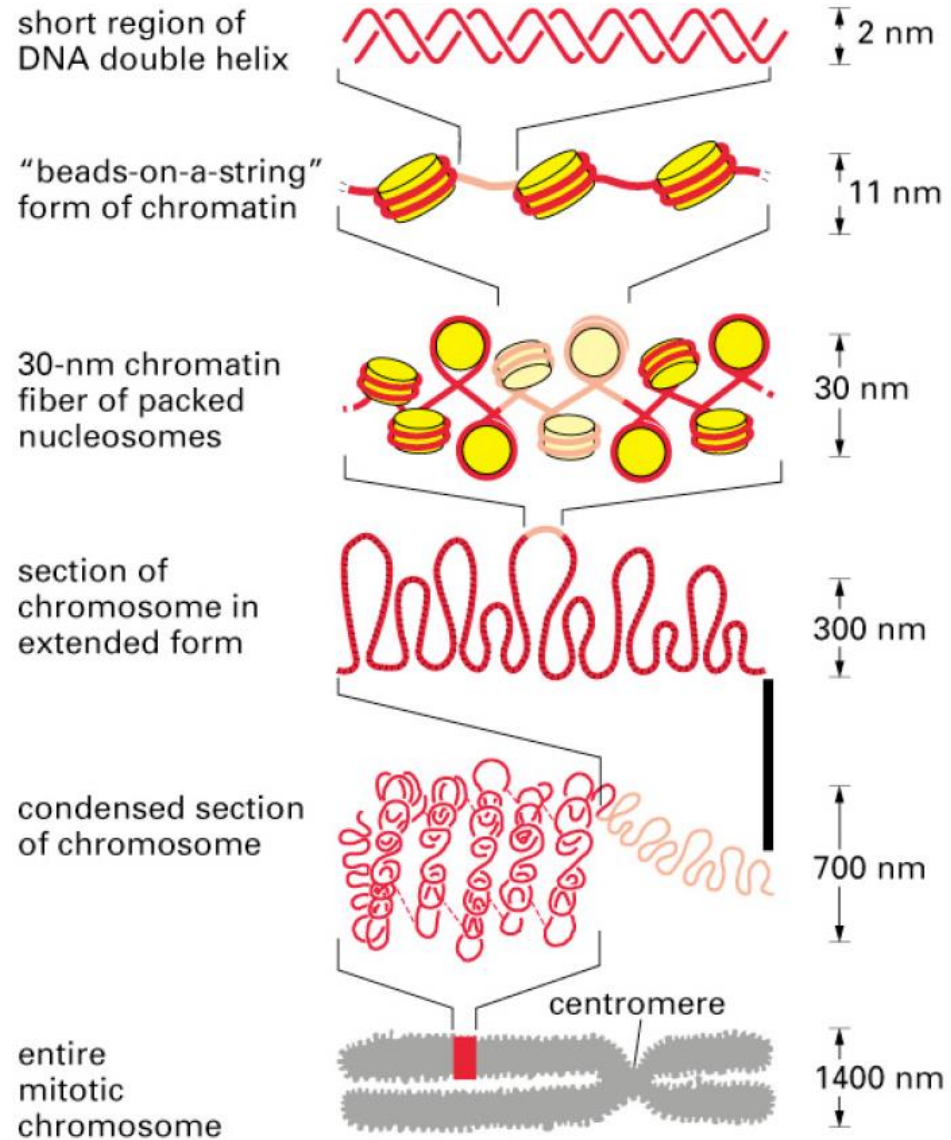


Fig. 4-55. Molecular Biology of the Cell (4th Ed.)





Nucleosome



- The nucleosome consists of 146 bp of DNA wrapped around a protein core of 8 histones

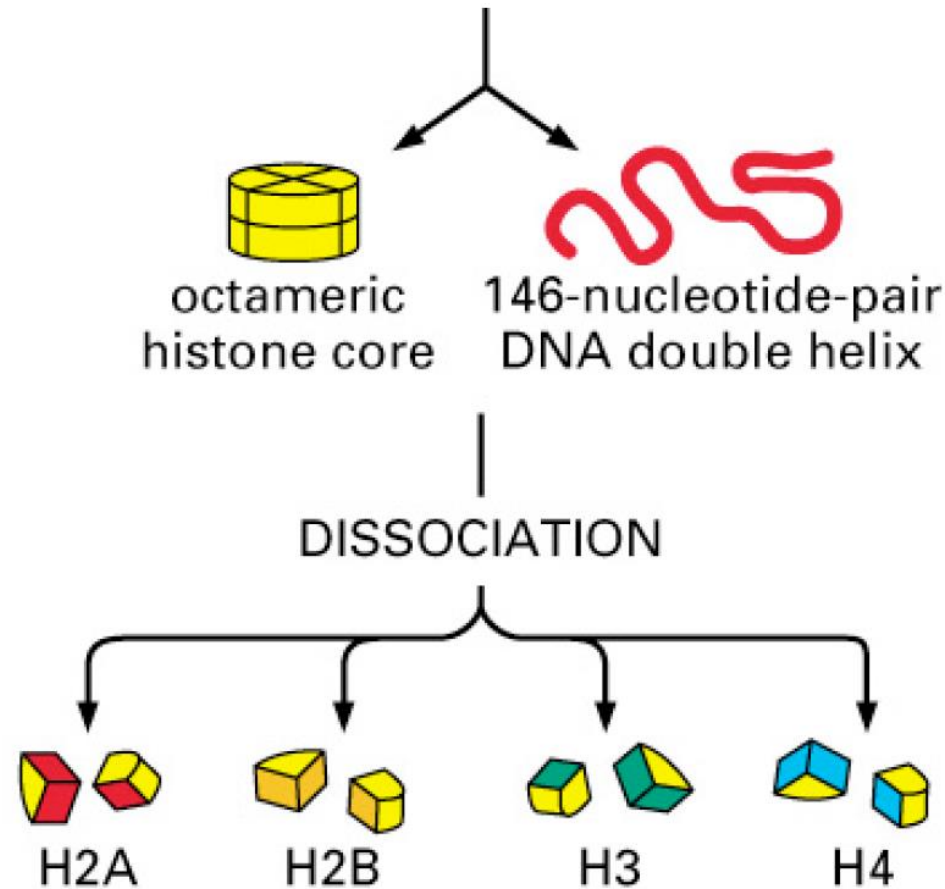


Fig. 4-24. Molecular Biology of the Cell (4th Ed.)



Other DNA binding proteins

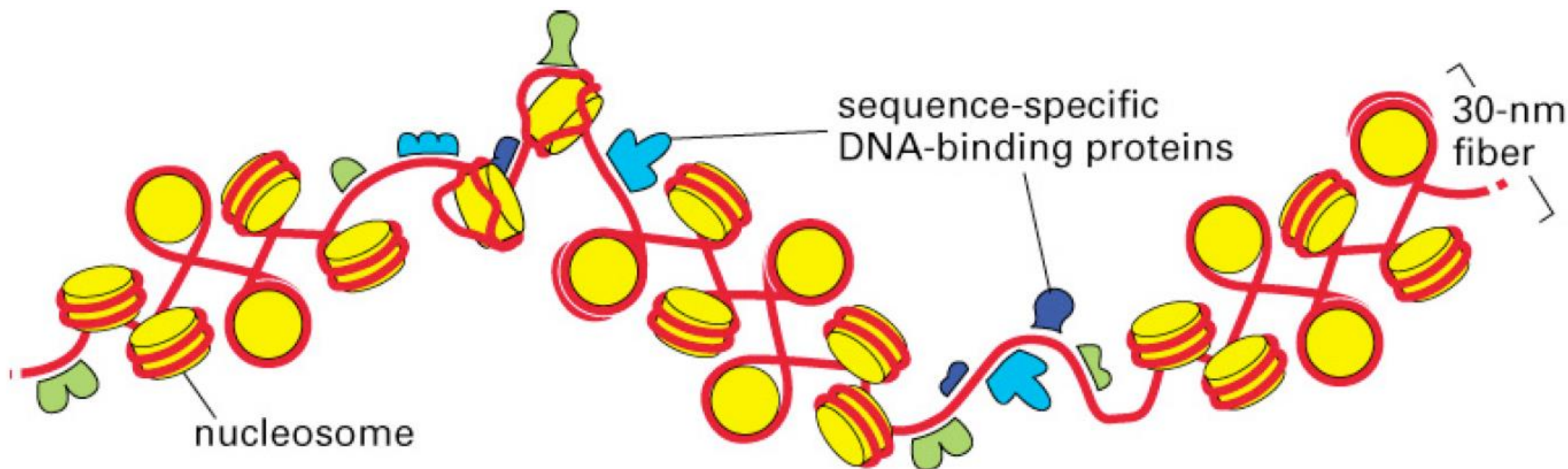


Fig. 4-30. Molecular Biology of the Cell (4th Ed.)





Chromatin loops



- The 30 nm fiber is organized into **loops** that can be opened up individually
- This allows individual genes and sets of genes to be accessed without a global unpacking of the chromosome

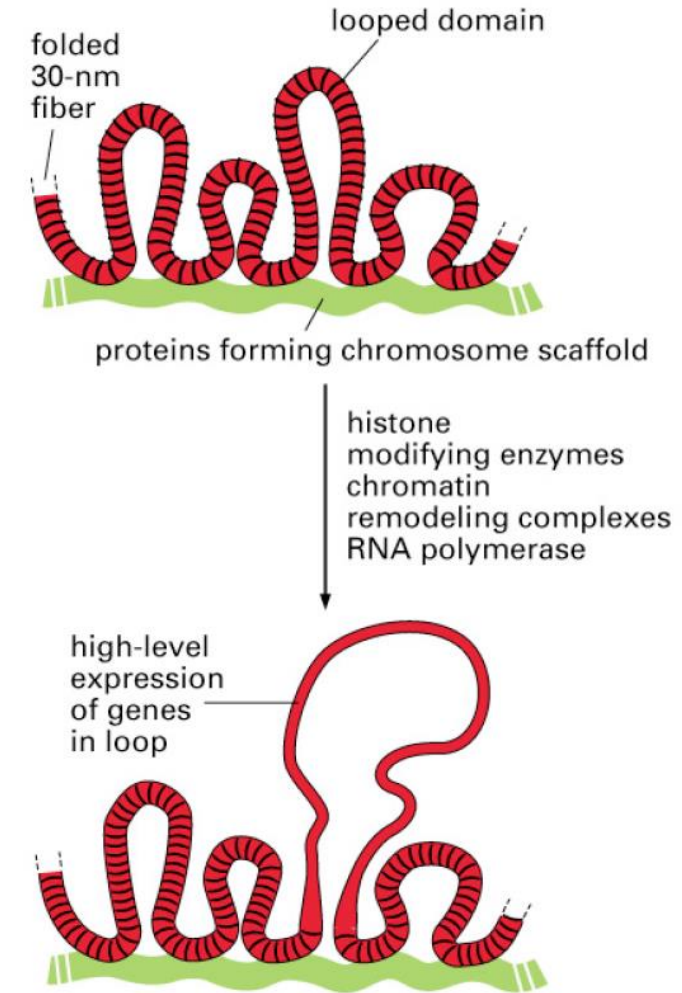


Fig. 4-44. Molecular Biology of the Cell (4th Ed.)





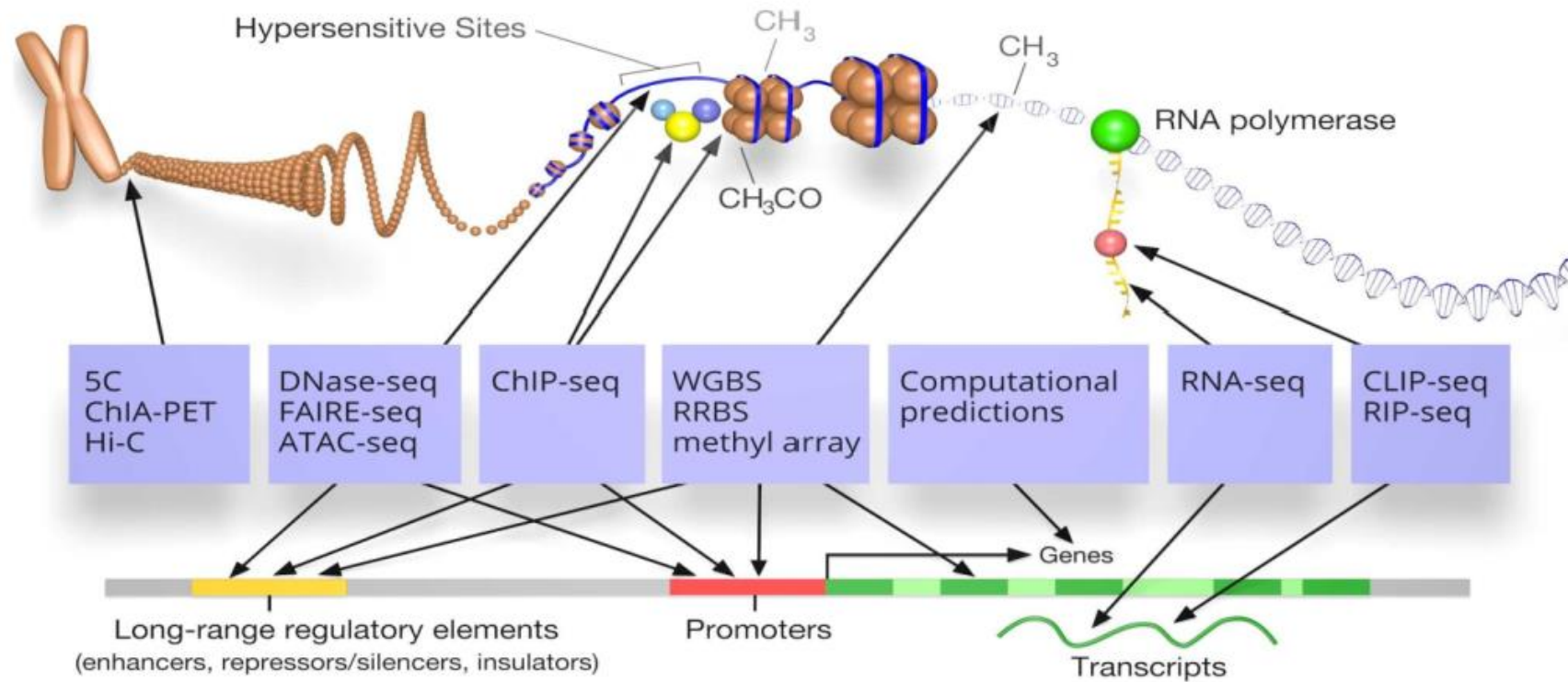
Main epigenetic mechanisms



- **DNA methylation**: Methyl group (an epigenetic factor found in some dietary sources) can tag DNA and activate or repress genes
- **Histone modification**: The binding of epigenetic factors to histone “tails” alters the extent to which DNA is wrapped around histones and the availability of genes in the DNA to be activated
- **Non-coding RNA (ncRNA)**: A functional RNA molecule that is transcribed from DNA but not translated into proteins. In general, ncRNAs function to regulate gene expression at the transcriptional and post-transcriptional levels



Sequencing for epigenetics



- Since DNA sequencing was first devised in 1977 by Frederick Sanger and his colleagues, sequencing methods have been evolved rapidly and volatile in sequencing objects (DNA, RNA, methylation, etc.)



ChIP-seq

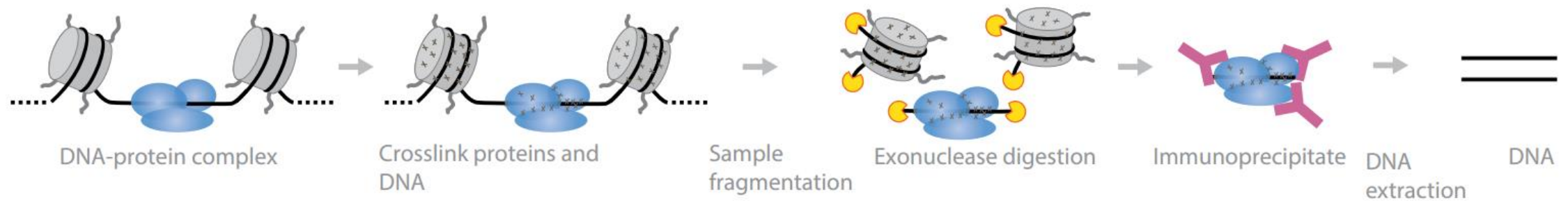




ChIP-seq



- **Chromatin immunoprecipitation sequencing (ChIP-seq)** is used primarily to determine how transcription factors and other chromatin-associated proteins influence phenotype-affecting mechanisms
- **Goal:** Identify both direct and indirect protein-DNA interactions
- **Experimental approach:** Targeted chromatin pulldown after crosslinking followed by sequencing of captured DNA fragments



Motivation for ChIP-seq



- General questions about gene (expression) regulation
 - Where are the relevant regulatory elements?
 - Which transcription factors (TFs) are involved?
 - What does chromatin state reveal?
 - What are the changes across conditions?
- To interpret and guide the studies of human genetics
 - Where are the relevant regulatory elements within the linkage block of GWAS signal or family studies?
 - What regulatory elements are likely to control critical developmental or phenotype-associated genes?
 - Guide resequencing and functional studies



• Pros

- Base-pair resolution of protein-binding site
- Can map specific regulatory factors or proteins
- The use of exonuclease eliminates the contamination by unbound DNAs

• Cons

- Nonspecific antibodies can dilute the pool of DNA-protein complexes of interest
- The target protein must be known and able to raise an antibody

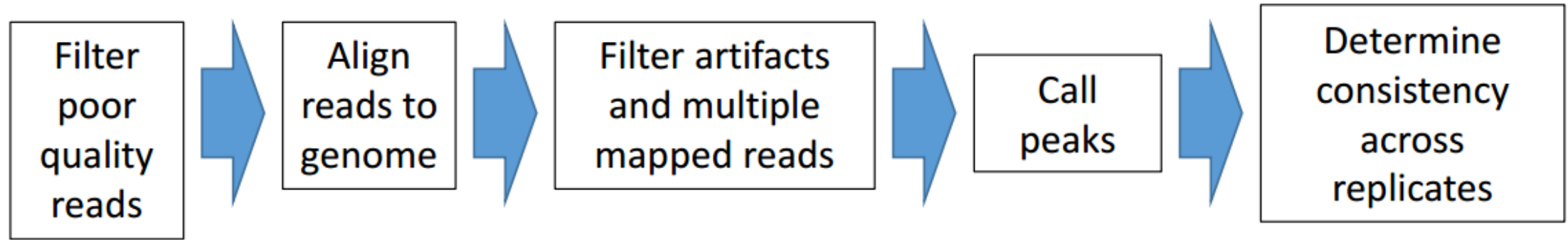


1. Map binding sites of a specific TF
2. Map interactions for other functional proteins
 - Co-activators (e.g. p300)
 - Structural proteins (e.g. CTCF)
 - Chromatin remodeling proteins (e.g. BRG1)
 - Nucleosome components (e.g. H2A)
3. Study chromatin regulation via histone modifications
 - **Annotation:** promoters, enhancers, gene bodies, etc.
 - **Function:** active enhancer, repressed chromatin, elongation, etc.





ChIP-seq data analysis pipelines

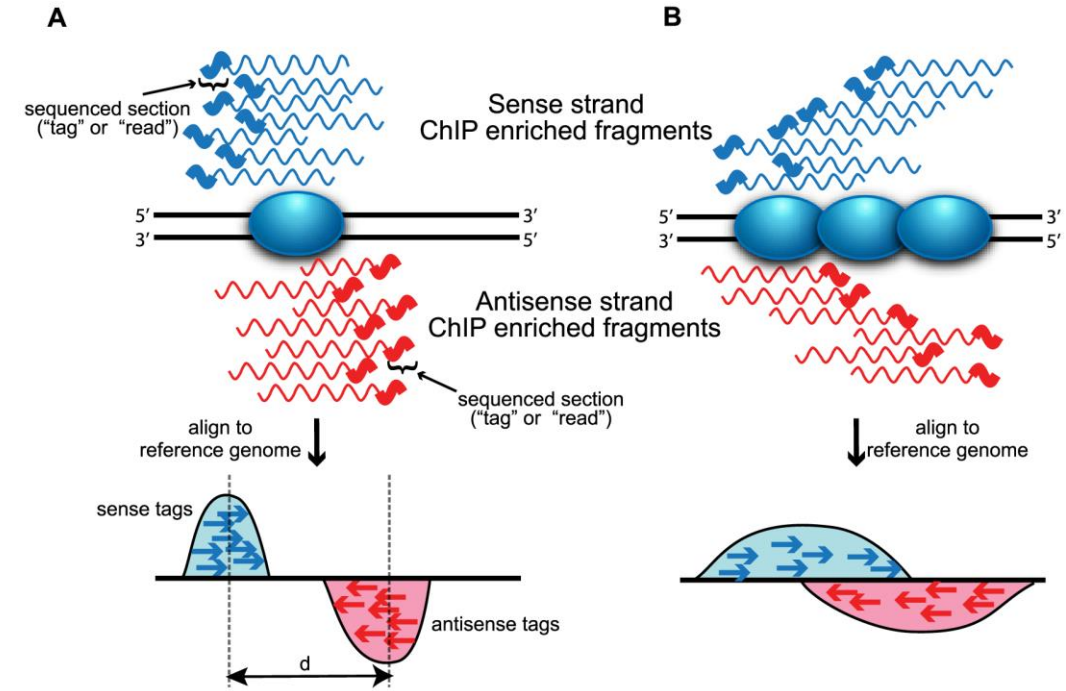


- ENCODE ChIP-seq pipeline: <https://github.com/ENCODE-DCC/chip-seq-pipeline2>
- UC Davis ChIP-seq pipeline: <https://github.com/kernco/chipseq-pipeline>

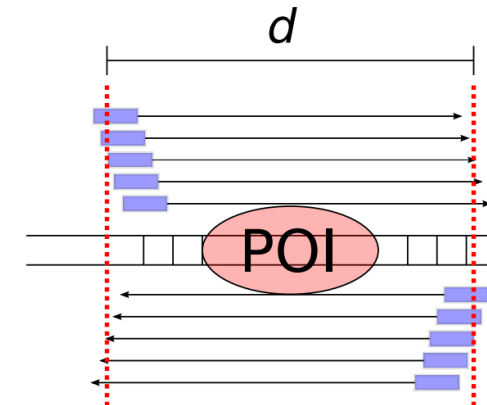


Peak calling

- **Peak calling algorithms** help define the protein-DNA binding sites by identifying regions where sequence reads are enriched in the genome after mapping
- Why does the **separation between peaks (d)** correspond to the **average sequenced fragment length**?
 - The purple box shows the region of the fragment that is actually sequenced (often 36bp)
 - The entire fragment is longer, with the exact size depending on the experimental fragmentation protocol
 - On average, the **protein of interest (POI)** is located in the middle of the fragment
 - Therefore, the average distance between reads corresponds to the average fragment length



The separation between peaks (d) corresponds to the average sequenced fragment length. Wilbanks et al. PLOS ONE, 2010

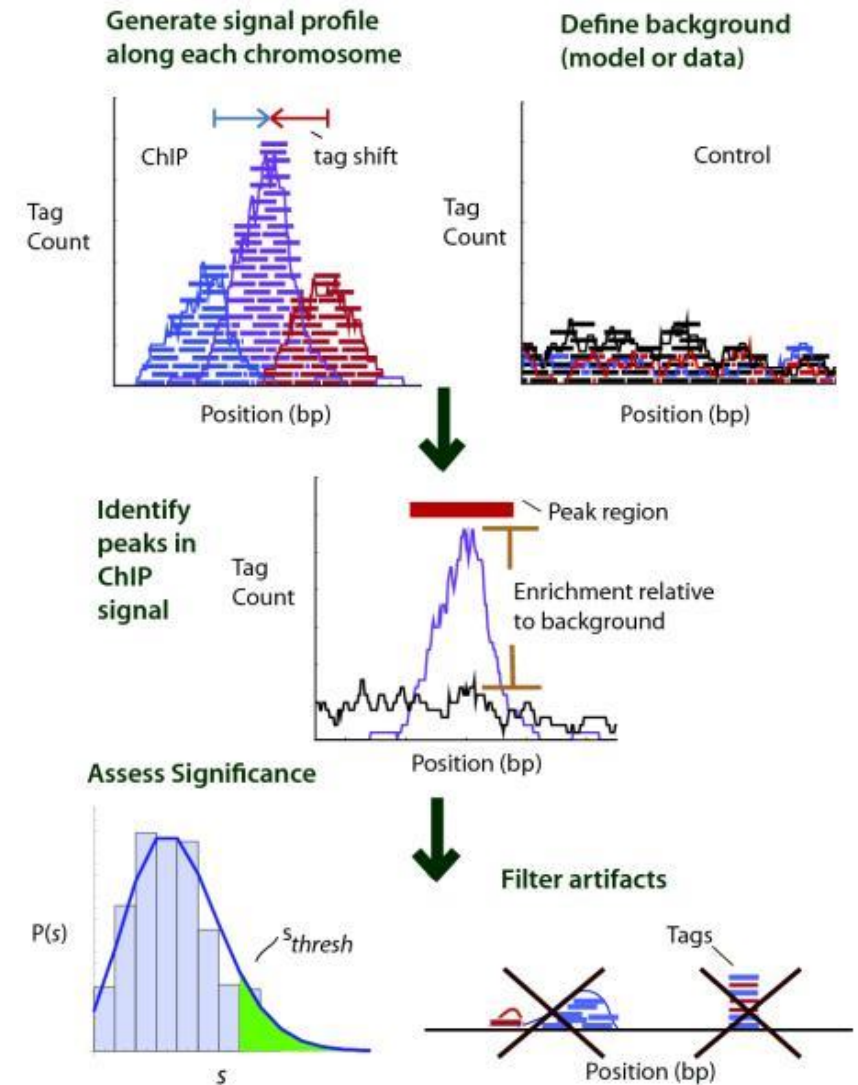




Peak evaluation



- Look for fold enrichment of the sample against input or expected background
- Estimate the significance of the fold enrichment using:
 - Poisson distribution
 - negative binomial (NB) distribution
 - background distribution from input DNA
 - model the background data to adjust for local variation (as in the MACS algorithm, described in Appendix A)



Pepke, S. et al., Nature Methods, 2009



ATAC-seq

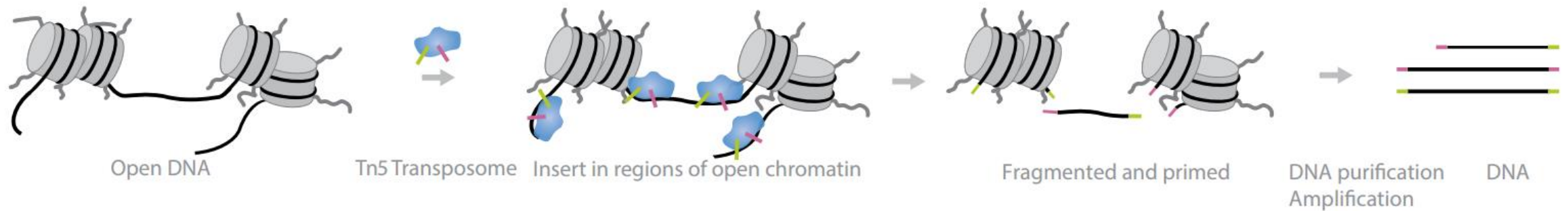




ATAC-seq



- **Assay for Transposase-Accessible Chromatin using sequencing (ATAC-seq):** is based on direct *in vitro* transposition of sequencing adapters into native chromatin – as a rapid and sensitive method for integrative epigenomics analysis ([Buenrostro, Jason D et al. Nature Method, 2013](#))
- **Experimental Approach:** DNA is incubated with Tn5 transposome, which performs adapter ligation and fragmentation of **open chromatin regions**. Deep sequencing of the purified regions provides base-pair resolution of **nucleosome-free regions** in the genome



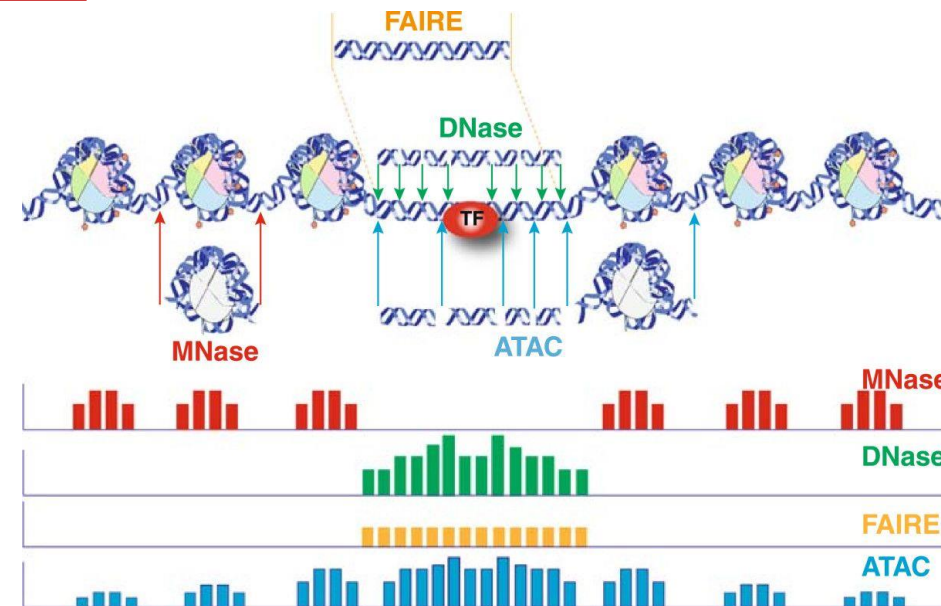


Compare with related techniques



Other techniques:

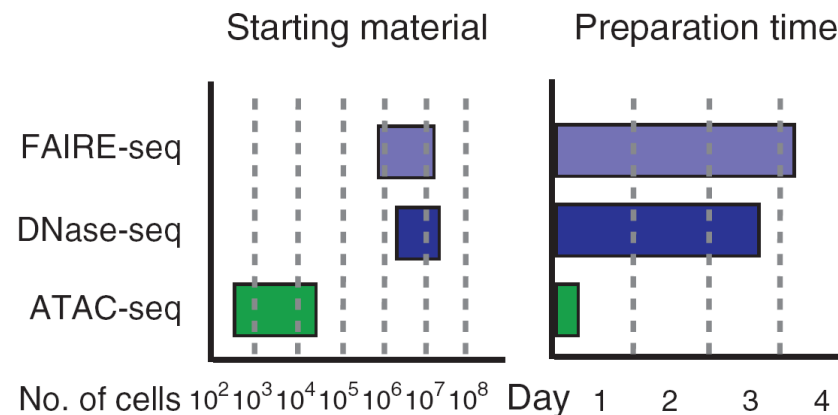
- **MNase-seq**: As MNase digests unprotected DNA, it is sensitive to open chromatin areas, and maps regions that are protected by nucleosomes (large open regions are digested)
- **DNase-seq**: As DNase I cuts within unprotected DNA, it maps open chromatin
- **FAIRE-seq**: Based on the phenol-chloroform separation of nucleosome-bound and free sonicated areas of a genome, it maps the open chromatin



Tsompana and Buck, Epigenetics & Chromatin, 2014

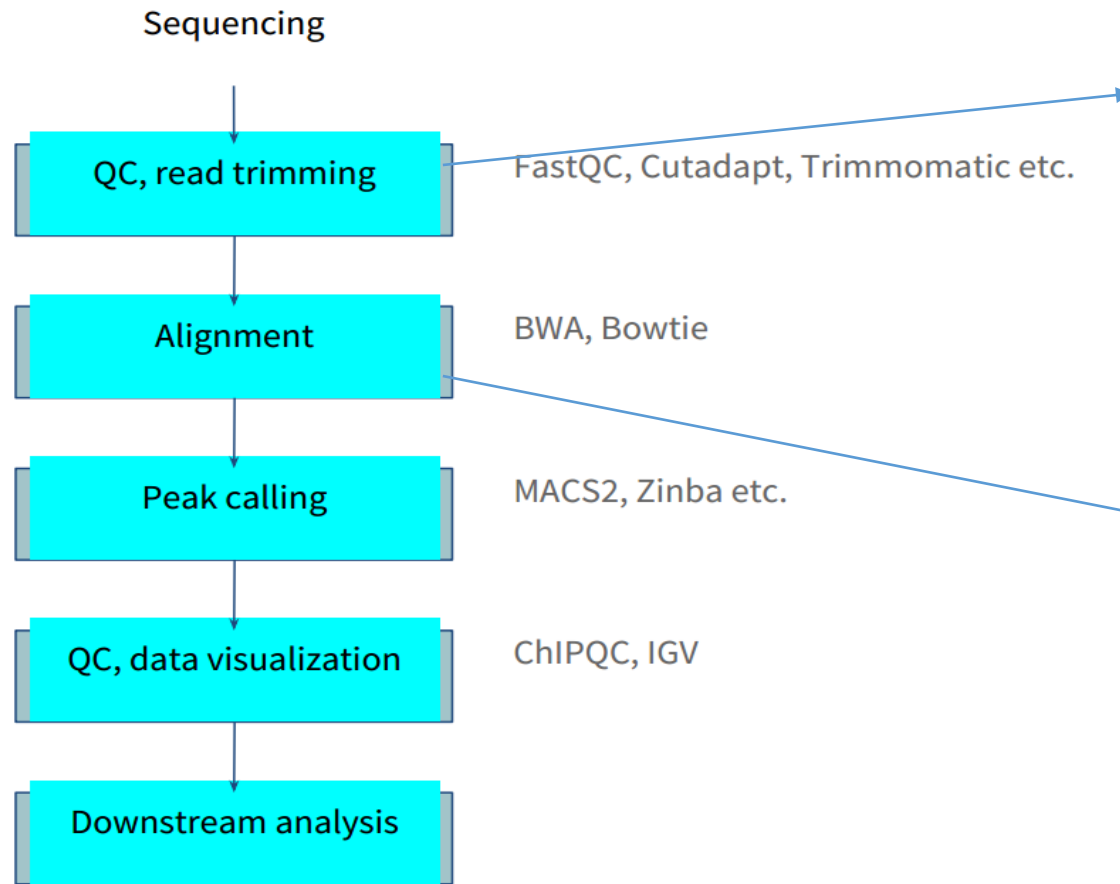
Pros of ATAC-seq

- Sample demand is greatly reduced
 - The amount of cells required is from 500 to 50,000
 - Compared to the sample requirements of 10^7 cells in other experiments, ATAC-seq is friendly to the small amount of cells
- Reducing sequencing depth
- One-time access to chromatin open regions on the whole genome
- Very high signal-to-noise ratio compared to FAIRE-seq



JD Buenrostro et al., Nature Methods, 2013

Workflow of ATAC-seq data processing



1. ATAC-seq can be further trimmed: estimating the percentage of sequence reads that map to the mitochondrial genome (the lower the better)
2. Use genome browsers to see raw tag density ---- UCSC, IGV, GIVE

1. **Full-read approach**
Bowtie2
Burrows-Wheeler Aligner (BWA)
2. **Chimeric alignment (unmapped from previous step, which may contains ligation junction)**
BWA
STAR
ChimeraScan



Hi-C

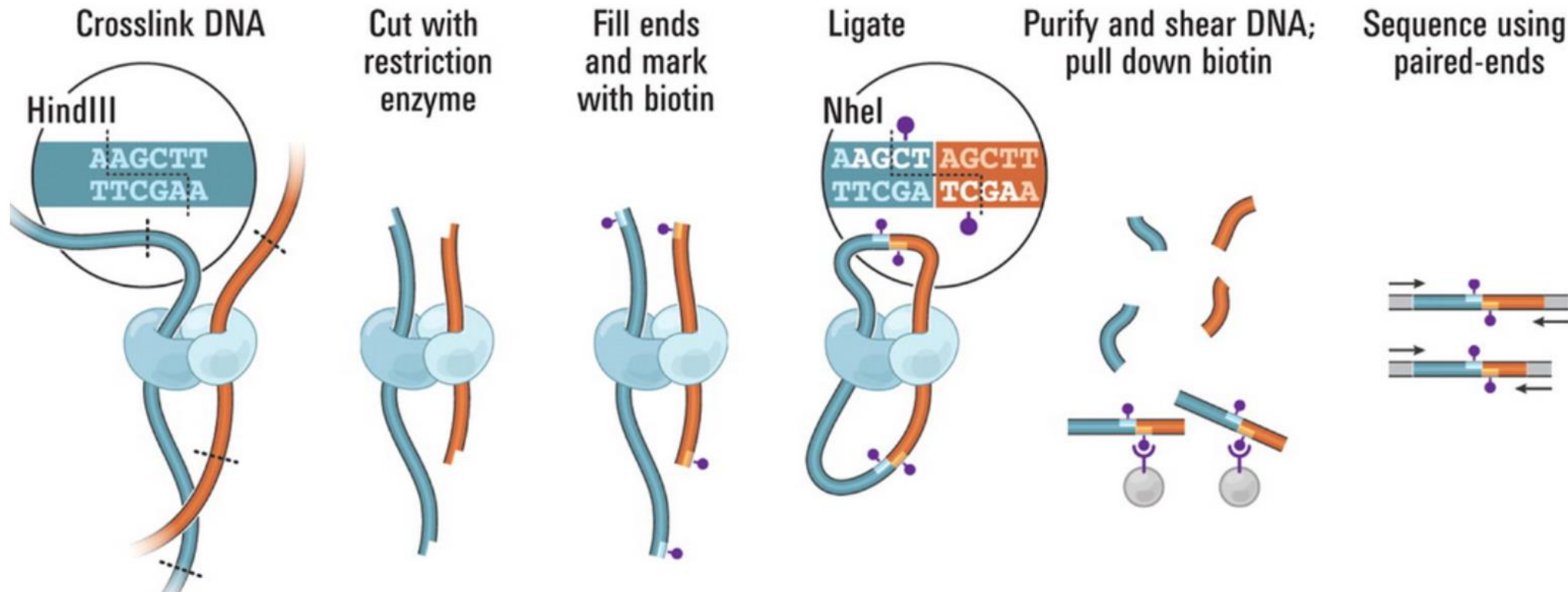




Hi-C



- Chromatin conformation capture sequencing (Hi-C) is used to analyze chromatin interactions (Lieberman-Aiden et al., Science, 2009; Rao et al., Cell, 2014)



Pros

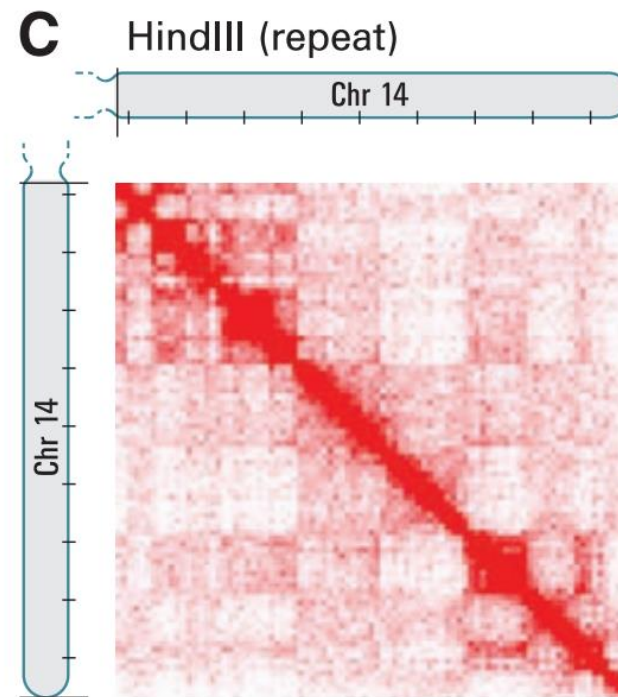
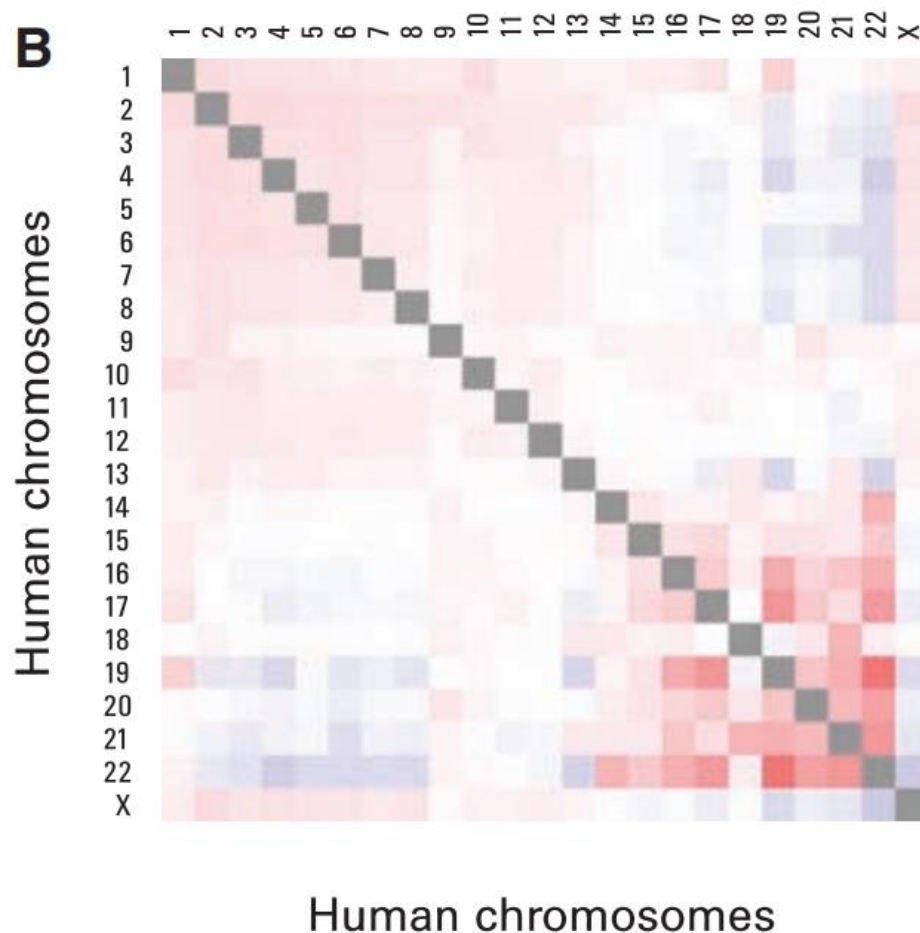
- Allows detection of long-range DNA interactions
- High-throughput method

Cons

- Detection may result from random chromosomal collisions
- Due to multiple steps, the method requires large amounts of starting material



What can we learn from Hi-C methods?



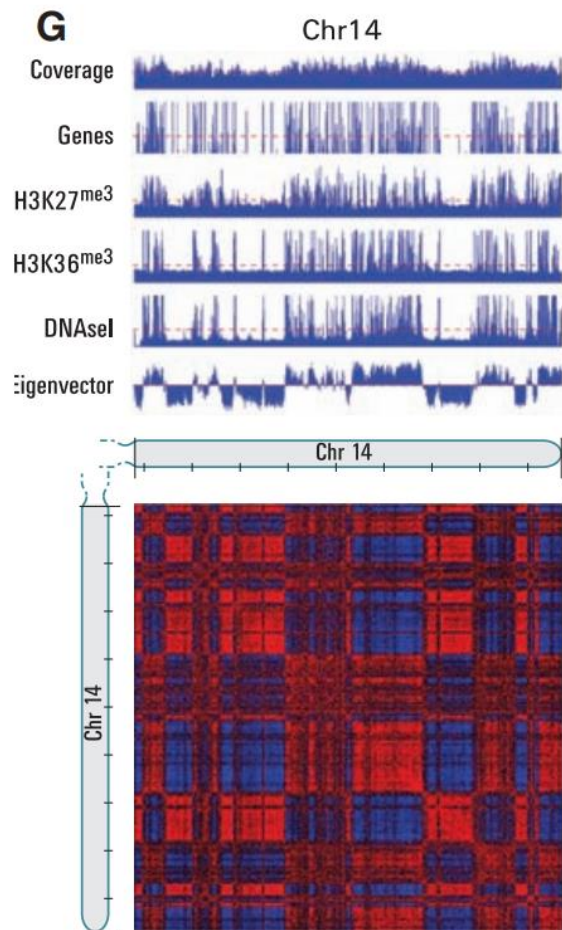
- Cross-linking frequencies between sequences genome-wide
- Often referred to as “interactions”
- Visualized in contact map

Lieberman-Aiden et al., Science, 2009

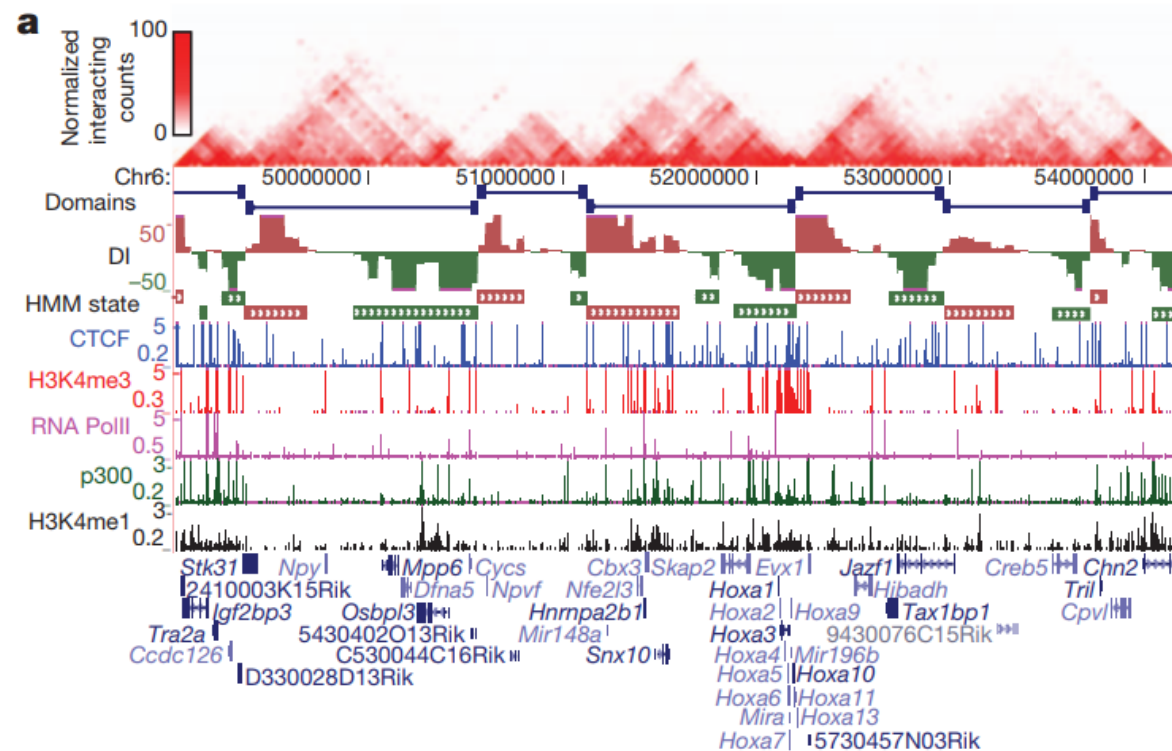
What can we learn from Hi-C methods?



- Chromosome compartmentalization



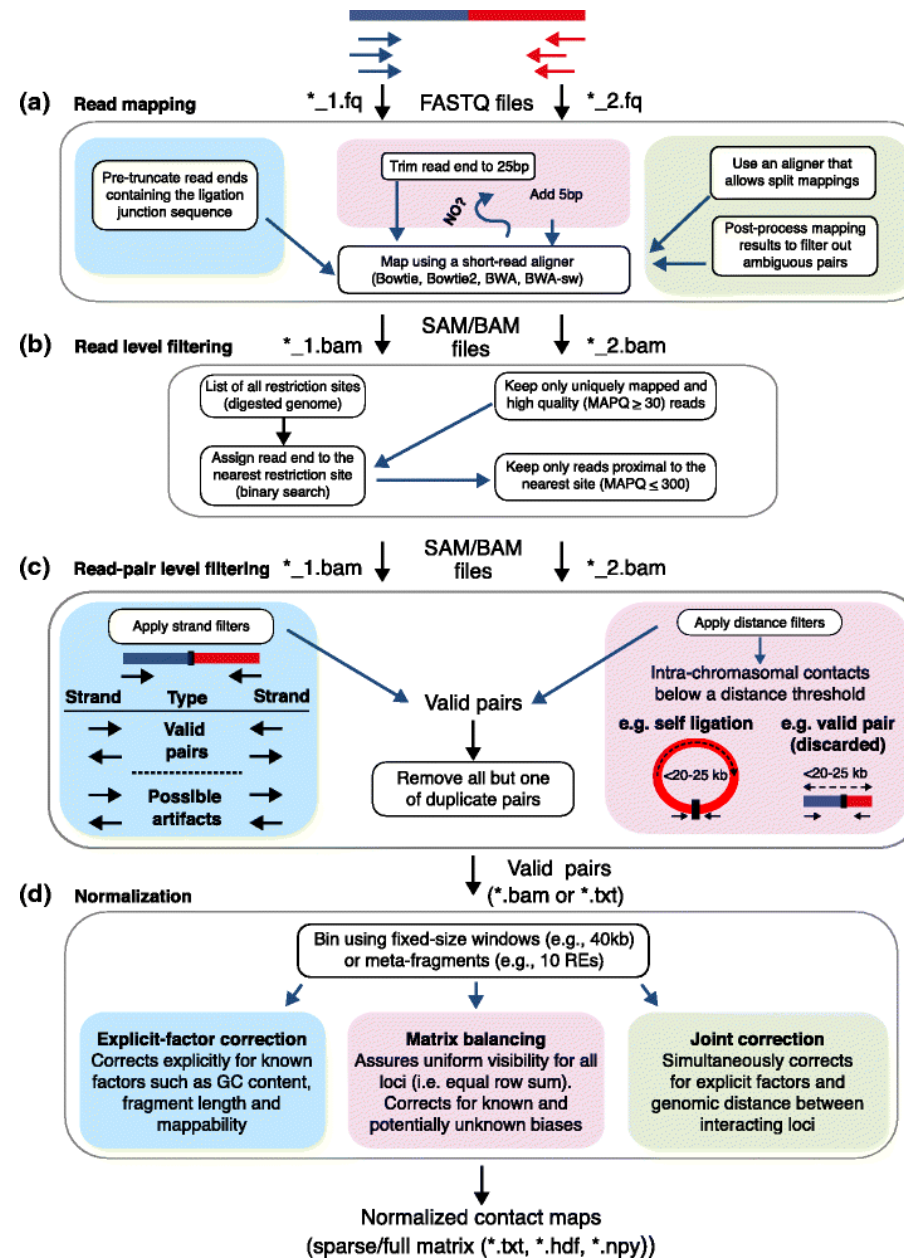
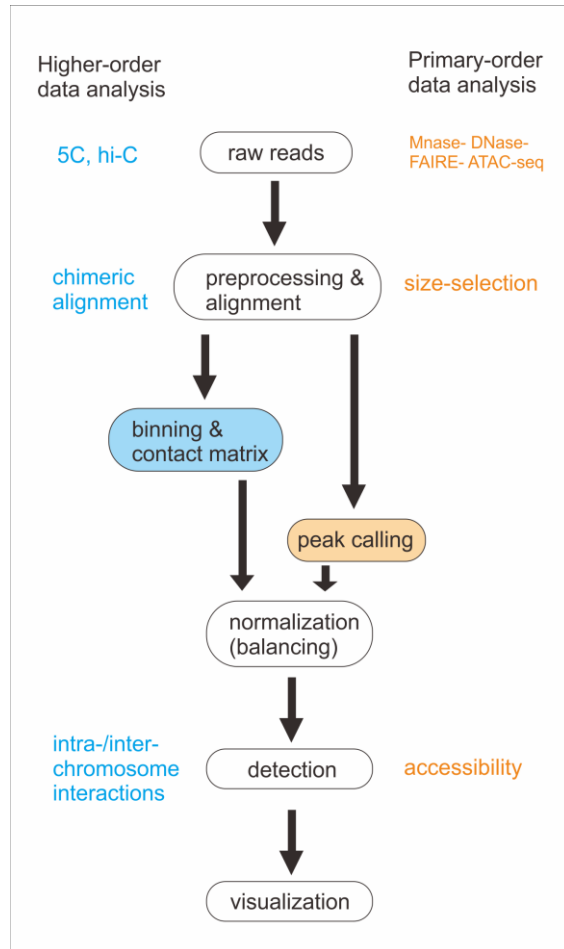
Lieberman-Aiden et al., Science, 2009



Dixon et al., Nature, 2012

Hi-C workflow

- Data analysis workflow for **higher-order** data and **primary-order** data
- The white box indicates common steps in the middle, and special analysis for different types are listed on two sides with colors



Overview of Hi-C analysis pipelines
Ay & Noble, Genome Biology, 2015



Software tools for Hi-C data analysis



Tool	Short-read aligner(s)	Mapping improvement	Read filtering	Read-pair filtering	Normalization	Visualization	Confidence estimation	Implementation language(s)
HiCUP [46]	Bowtie/Bowtie2	Pre-truncation	✓	✓	—	—	—	Perl, R
Hiclib [47]	Bowtie2	Iterative	✓ ^a	✓	Matrix balancing	✓	—	Python
HiC-inspector [131]	Bowtie	—	✓	✓	—	✓	—	Perl, R
HIPPIE [132]	STAR	✓ ^b	✓	✓	—	—	—	Python, Perl, R
HiC-Box [133]	Bowtie2	—	✓	✓	Matrix balancing	✓	—	Python
HiCdat [122]	Subread	— ^c	✓	✓	Three options ^d	✓	—	C++, R
HiC-Pro [134]	Bowtie2	Trimming	✓	✓	Matrix balancing	—	—	Python, R
TADbit [120]	GEM	Iterative	✓	✓	Matrix balancing	✓	—	Python
HOMER [62]	—	—	✓	✓	Two options ^e	✓	✓	Perl, R, Java
Hicpipe [54]	—	—	—	—	Explicit-factor	—	—	Perl, R, C++
HiBrowse [69]	—	—	—	—	—	✓	✓	Web-based
Hi-Corrector [57]	—	—	—	—	Matrix balancing	—	—	ANSI C
GOTHIC [135]	—	—	✓	✓	—	—	✓	R
HiTC [121]	—	—	—	—	Two options ^f	✓	✓	R
chromoR [59]	—	—	—	—	Variance stabilization	—	—	R
HiFive [136]	—	—	✓	✓	Three options ^g	✓	—	Python
Fit-Hi-C [20]	—	—	—	—	—	✓	✓	Python





Proteomics



■ What is proteomics?



- The **proteome** is the entire set of proteins that is produced or modified by an organism or system
 - The term of “proteome” was coined in 1994 by Marc Wilkins, blending “protein” and “genome”
- **Proteomics** is the large-scale study of proteins
 - The term was coined in 1997, in analogy to “genomics”
 - Proteomics has enabled the identification of ever increasing numbers of proteins
 - It is an interdisciplinary domain that covers all levels of proteins, e.g. protein composition, structure and activity
 - Important for functional genomics and systems biology





Approaches to protein identification



- There are 3 main approaches to protein identification
 - Direct protein sequencing
 - Gel electrophoresis
 - Mass spectrometry
- High-throughput protein sequencing is to deduce the amino acid sequence of a protein. It is still very difficult
- Currently, research is focused on **peptide sequencing**, i.e. getting the amino acid sequence of a short fragment of a protein (of length ≈ 10)





Mass spectrometry

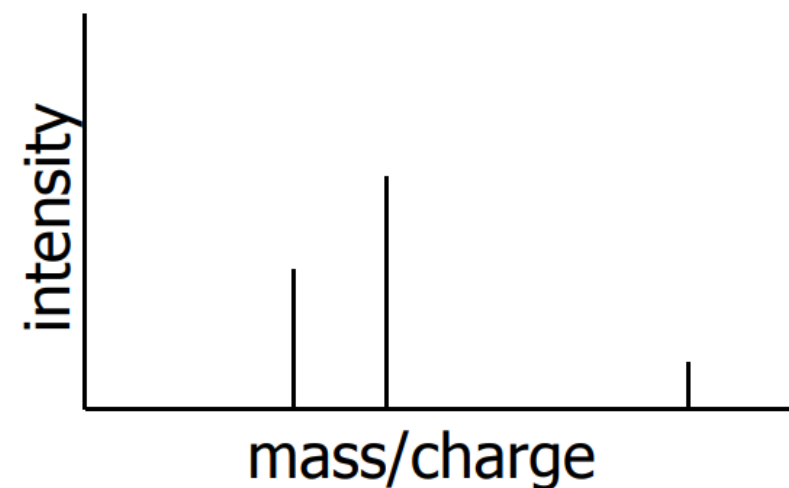


- Key for deducing the peptide sequence: mass
- **Mass spectrometry** is an analytical technique that can separate and measure samples with different mass/charge ratios
- Example:

Sample 1: $m/z=100\text{Da}$, 10mol

Sample 2: $m/z=50\text{Da}$, 50mol

Sample 3: $m/z=33\text{Da}$, 30mol



Dalton(Da) is a mass unit. E.g. H is of mass 1Da





History



- Peptide sequencing was invented by Pehr Edman (1949) and Frederick Sanger (1955)
- In 1966, Biemann et al. successfully sequenced a peptide using a mass spectrometer machine
- During 1980s, sequencing using mass spectrometry becomes popular



Amino acid residue mass



- Observations from the table:
 - Amino acid residue = amino acid losing a water
 - **I** and **L** have the same mass
 - Smallest mass is **G** (57.05 Da)
 - Largest mass is **W** (186.21 Da)
 - Most of the amino acids have different masses. Thus, with high chance, different peptides have different masses
- The mass given by a mass spectrometer has a max. error $\pm 0.5\text{Da}$. It can separate most of the peptides

A	71.08		M	131.19
C	103.14		N	114.1
D	115.09		P	97.12
E	129.12		Q	128.13
F	147.18		R	156.19
G	57.05		S	87.08
H	137.14		T	101.1
I	113.16		V	99.13
K	128.17		W	186.21
L	113.16		Y	163.18





Protein identification process



- **Input:** A protein sample

A) **Biology part:**

- 1. Digest the protein into a set of peptides
- 2. By HPLC + Mass Spectrometer, separate the peptides
- 3. Select a particular peptide
- 4. Fragment the selected peptide
- 5. Get the tandem mass (MS/MS) spectrum of the selected peptide

B) **Computing part:**

- De novo peptide sequencing
- Protein database search



Digest a protein into peptides

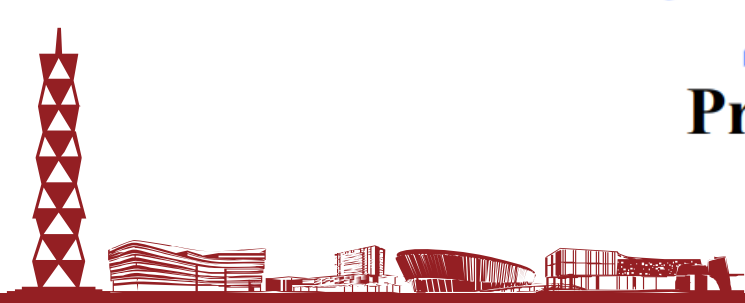
- By an enzyme, digest a protein into short peptides
- If we digest a protein using **trypsin**,
 - It digests the protein at K or R that are **not followed by P**
 - After digestion, we will get a set of peptides ending with K or R
- E.g. ACCHCKCCV**R**PPC**R**CA -> ACCHCK, CCVRPPCR



Proteins



Peptides

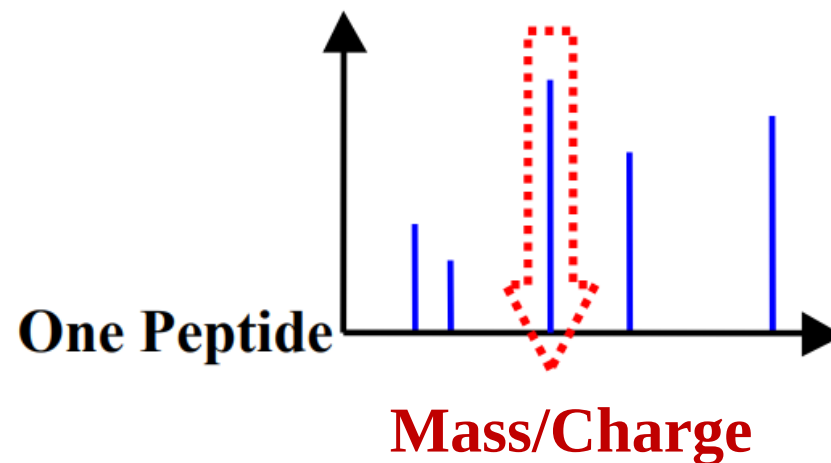




Selecting a particular peptide



- **HPLC** (High Performance Liquid Chromatography) can separate a set of peptides in a high pressure liquid chromatography
- After HPLC, the mixture of peptides are analyzed by MS. Then, we get the MS spectrum



- The peptide of a particular mass is selected





Fragmentation of peptide (I)

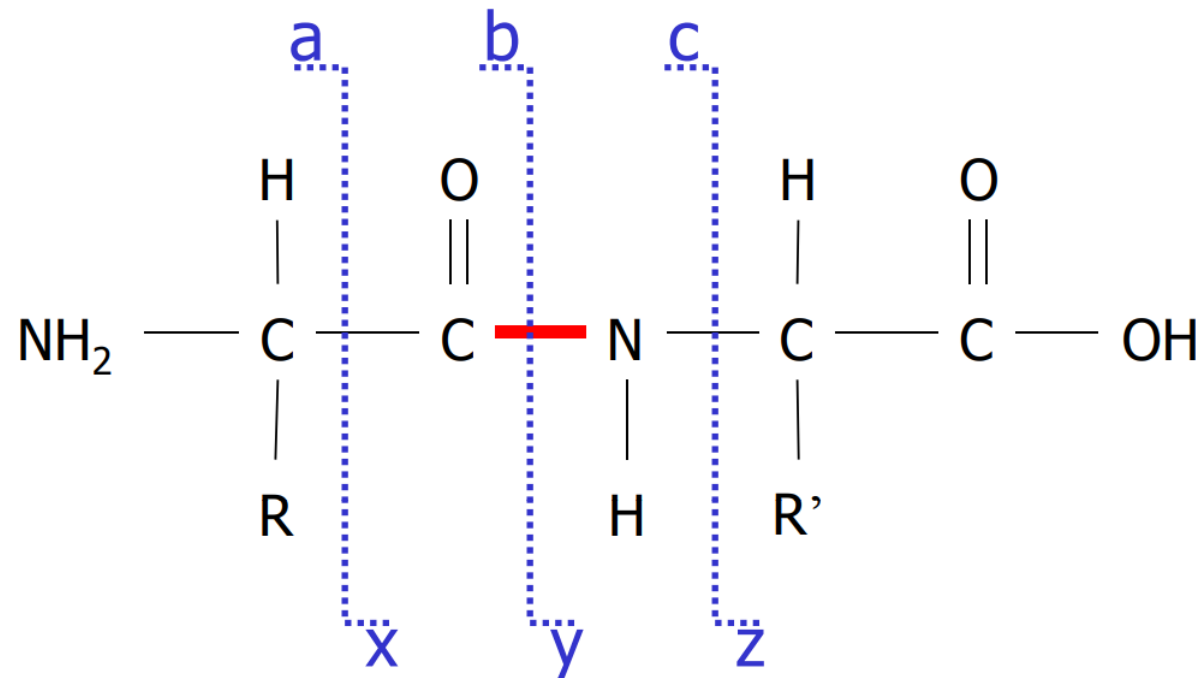


- Fragmentation tries to break the selected peptide at all positions in the peptide background
- Usually, fragmentation is by **Collision Induced Dissociation (CID)**
 - The peptide is passed into the collision cell, which has been pressurized with **argon** (i.e. inert gas)
 - Collision between peptide and argon break the peptide
- Each peptide is usually fragmented into 2 pieces
 - Prefix fragment and suffix fragment (either one fragment will be charged but not both)



Fragmentation of peptide (II)

- Most often, the peptide is broken at C-C, C-N, N-C bonds
 - Result in a-ions, b-ions, c-ions, x-ions, y-ions, and z-ions
- Based on experiment
 - The intensity of y-ions is larger than that of b-ions
 - The intensities of other ions are even smaller



Fragmentation of peptide (III)

$$r = w(\text{CTVFT})$$

$$w = w(\text{CTVFTEPREFK})$$

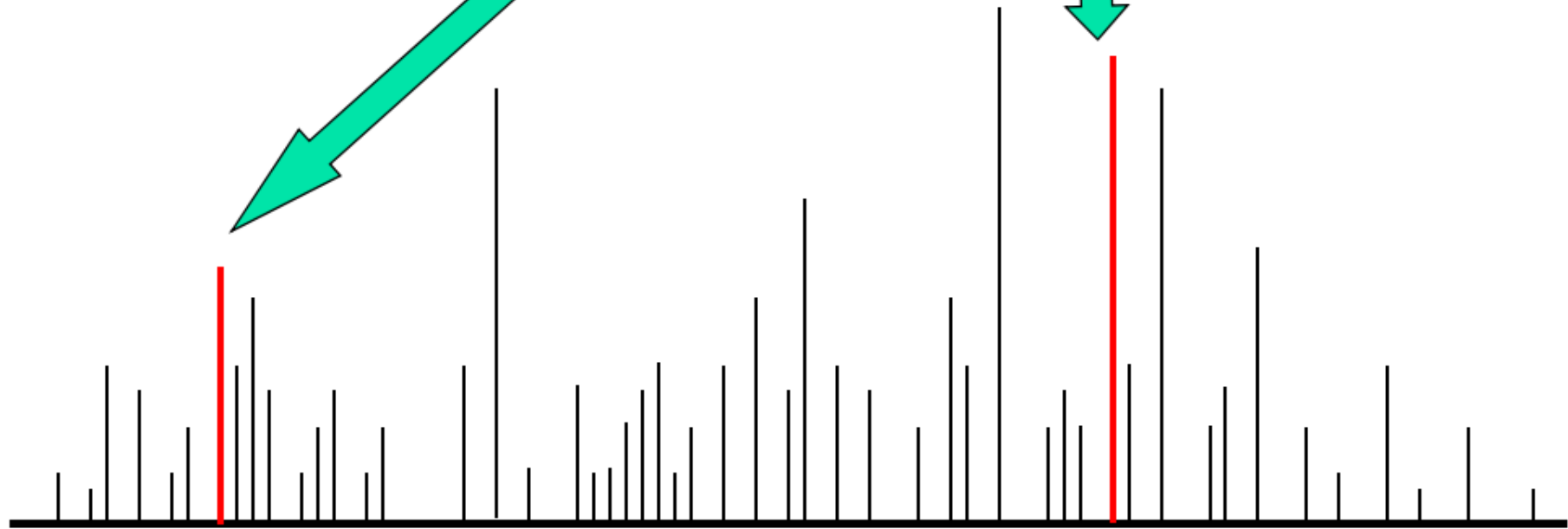
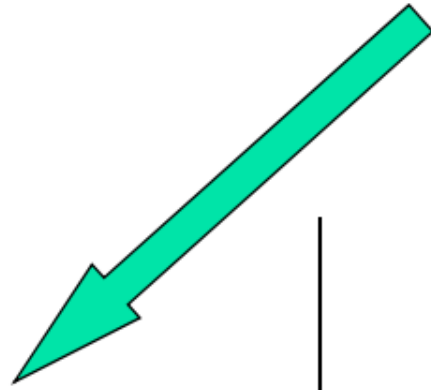
CTVFTEPREFK



fragmentation

CTVFT

EPREFK

 $r+1$ (mass of b-ion) $w-r+19$ (mass of y-ion)

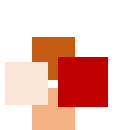


Mass of the ions (I)



- Let A be the set of amino acids. For every $a \in A$, $w(a)$ = mass of its residue
- Let $P = a_1a_2...a_k$ be a peptide
 - $w(P) = \sum_{1 \leq j \leq k} w(a_j)$
- Actual mass of the peptide with sequence P is
 - $w(P) + 18$ (since it has an extra H_2O)
- Mass of b-ion of the first i amino acids is
 - $b_i = 1 + w(a_1a_2...a_i)$
- Mass of y-ion of the last $(k - i)$ amino acids is
 - $y_{i+1} = 19 + w(a_{i+1}...a_k)$
- Note: $b_i + y_{i+1} = 20 + w(P)$





Mass of the ions (II)



- E.g. a peptide $P = \mathbf{SAG}$
 - $w(P) = w(S) + w(A) + w(G) = 215.21$
 - Actual mass of $P = w(P) + 18 = 233.21$
 - $y_1 = w(SAG) + 19 = 234.21$
 - $y_2 = w(AG) + 19 = 147.13$
 - $y_3 = w(G) + 19 = 76.05$
 - $b_1 = w(S) + 1 = 88.08$
 - $b_2 = w(SA) + 1 = 159.16$
 - $b_3 = w(SAG) + 1 = 216.21$





Other ion types



- Apart from a-ion, b-ion, c-ion, x-ion, y-ion, and z-ion, there are also variations with additional loss of
 - A water molecule
 - An ammonia molecule
 - A water and an ammonia molecule
 - Two water molecules
- E.g. $y\text{-H}_2\text{O}$, $y\text{-NH}_3$, $y\text{-H}_2\text{O-H}_2\text{O}$, $y\text{-H}_2\text{O-NH}_3$



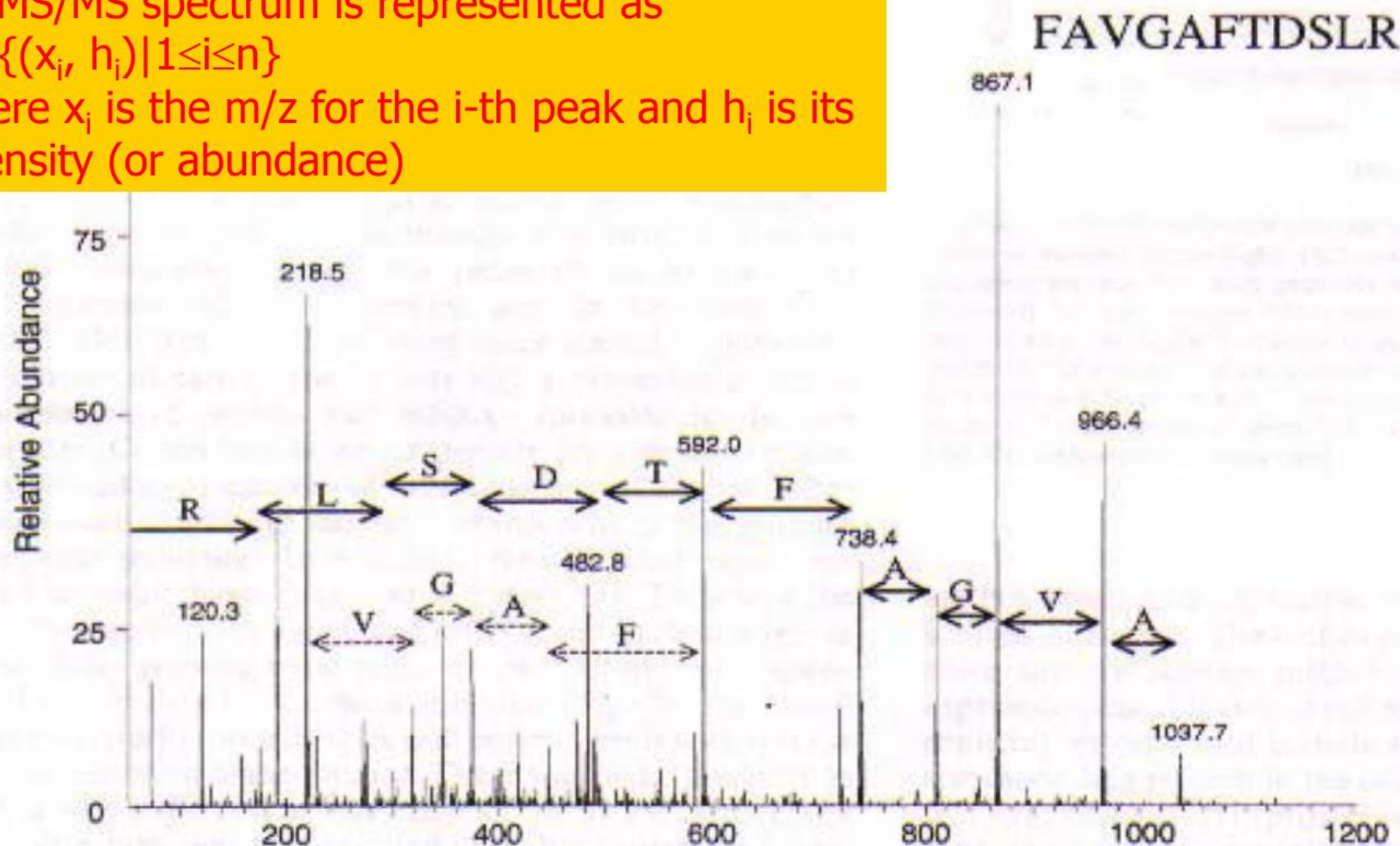
Tandem mass spectrum (MS/MS Spectrum)



An MS/MS spectrum is represented as

$$M = \{(x_i, h_i) | 1 \leq i \leq n\}$$

where x_i is the m/z for the i -th peak and h_i is its intensity (or abundance)



- There are mainly 3 computational problems
 - 1. De novo peptide sequencing (Appendix B)
 - 2. Peptide identification through database search (Appendix C)
 - 3. Identification of PTM (post-translational modification)





Online resources for proteomics



- ExPASy at SIB (Swiss Institute of Bioinformatics):
 - <https://expasy.org>
 - A Bioinformatics Resource Portal providing access to scientific databases and software tools in different areas of life science
 - Proteomics is a foremost category (<https://www.expasy.org/proteomics>)
- PRIDE at EBI: A database for mass spectrometry
 - <http://www.ebi.ac.uk/pride/>
 - A centralized, standards compliant, public data repository, including information on protein and peptide identifications, PTMs, and supporting spectral evidence
- Other repositories:
 - Global Proteome Machine Database (GPMDB): (<http://gpmdb.thegpm.org/index.html>)
 - PeptideAtlas (<http://www.peptideatlas.org>)
 - MOPED (Model Organism Protein Expression Database): (<http://moped.proteinspire.org>)
 - Tranche (<http://www.proteomexchange.org/databases/tranche>)





Metabolomics



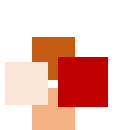
■ Metabolomics defined



- **Metabolomics** is the “systematic study of unique chemical fingerprints that specific cellular processes leave behind” ¹
- Generally speaking, metabolomics is limited to molecules 1.5 kDa (1500 g/mol) or less
 - Metabolomics is sometimes referred to as “small molecule profiling”
- Remember that a metabolic profile represents but a single snapshot in time. The metabolome is a dynamic entity and change from second to second
 - More recent efforts focus on **Real Time Metabolomics** which can measure hundreds of metabolites in a living cell in real time

¹ Davis, B (April 2005). "Growing pains for metabolomics". *The Scientist*. **19** (8): 25–28.





What molecules are measured in Metabolomics?



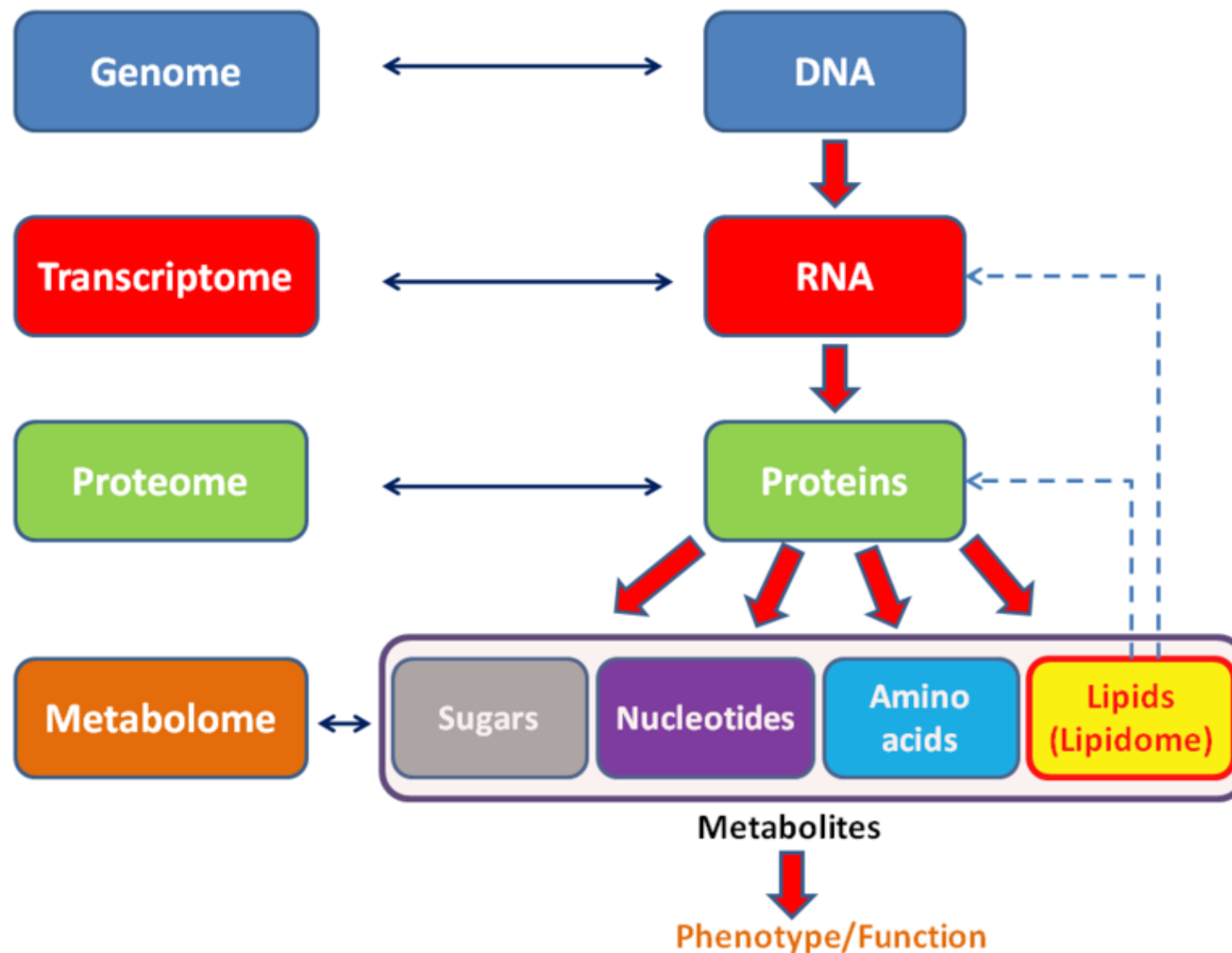
上海科技大学
ShanghaiTech University

- **Endogenous** (produced **within** the organism): amino acids, lipids, cofactors, nucleotides, sugars, hormones, etc.
- **Exogenous** (produced **external** to the organism): drugs, toxins, environmental contaminants, pesticides, herbicides, etc.



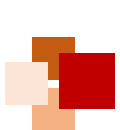


A Geneticist's view of Metabolomics



<https://en.wikipedia.org/wiki/User:Lmaps>





Relationships among the omics



- *Genome* can tell what could happen,
- *Transcriptome* can tell what appears to be happening,
- *Proteome* can tell what makes it happen, and
- *Metabolome* can tell what has happened and what is happening

Dettmer K. et al (2007). "Mass spectrometry-based metabolomics". Mass Spectrometry Reviews. **26**(1): 51–78.

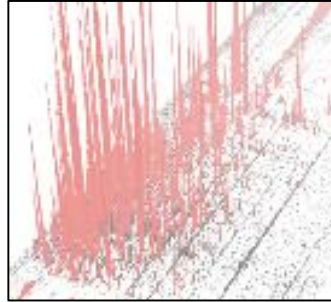




Metabolomics: Two basic approaches

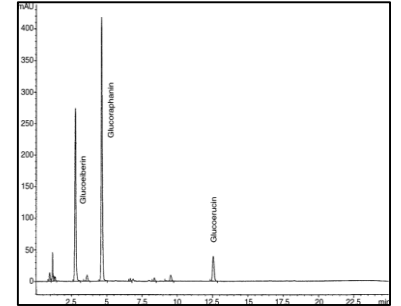


Untargeted: Shotgun Profiling



- Provides the opportunity to see everything
 - Many known, and even more unknowns
 - Relative abundances of hundreds of compounds
 - Data analysis is more difficult
- Potentially a powerful phenotyping tool
 - Need to have differences between bio samples for GWAS/QTL mapping

Targeted: Classical Analytical Biochemistry



- Can optimize conditions for specific metabolites
 - Flavonoids, carotenoids, hormones, etc.
 - Can be quantified
 - Data analysis is easier
- Downside: standards are needed
 - The number of known compounds is limited





Bio samples vary greatly as to how many metabolites they contain



Bio Sample	Total Number of Metabolites Detected	Percent Quantified
Sweat	91	86%
Cerebrospinal Fluid	445	99.5%
Saliva	1245	71%
Urine	2603	68%
Blood	3678	84%
Feces	6810	26%

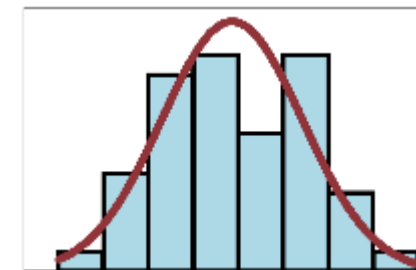




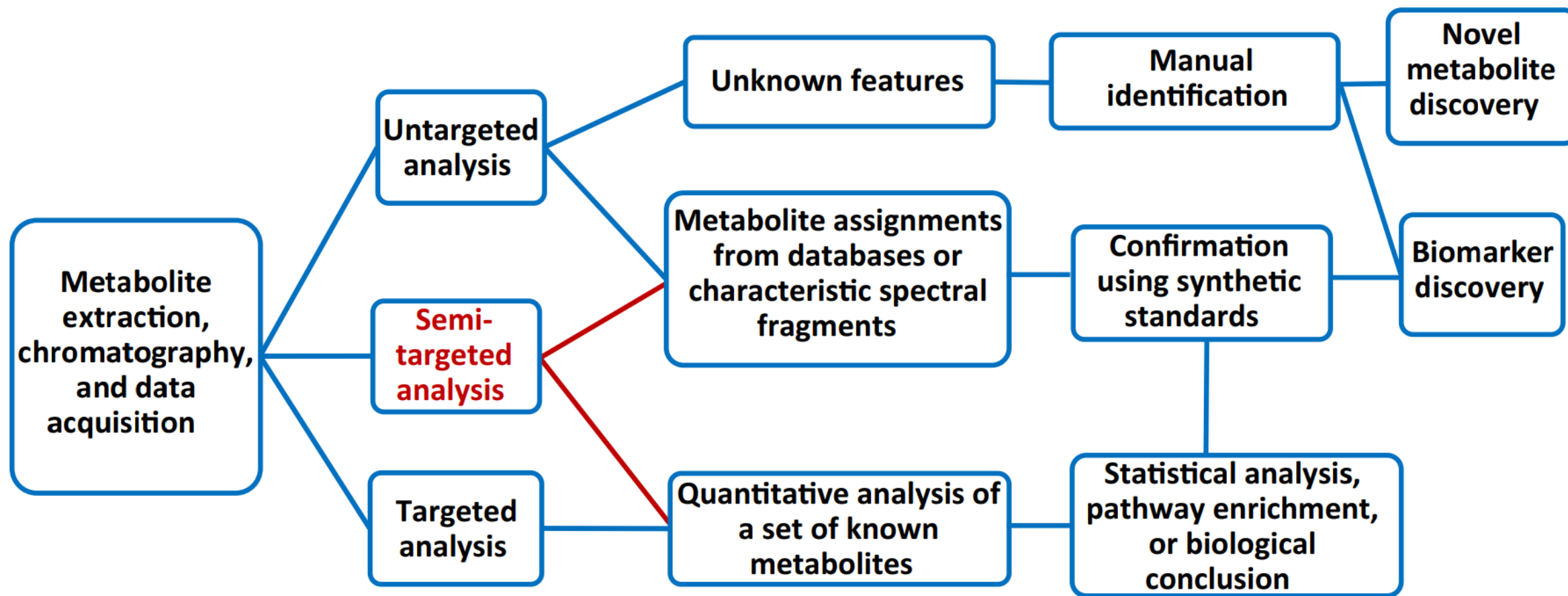
Statistical and sampling considerations



- Experimental design is critical
 - 3 repeats minimum, 5 repeats preferred
 - Sample pooling is beneficial
- Biological are preferred over technical repeats
 - In general, biological variance exceeds technical variance
- Many free software packages are available
 - PCA is often used in untargeted metabolic profiling
 - Machine learning methods can improve metabolite identification
- Substantial metabolite variation is normal even in carefully collected samples
 - Not all metabolites are tightly regulated. This can hamper statistical analysis



General schema of metabolomics workflows

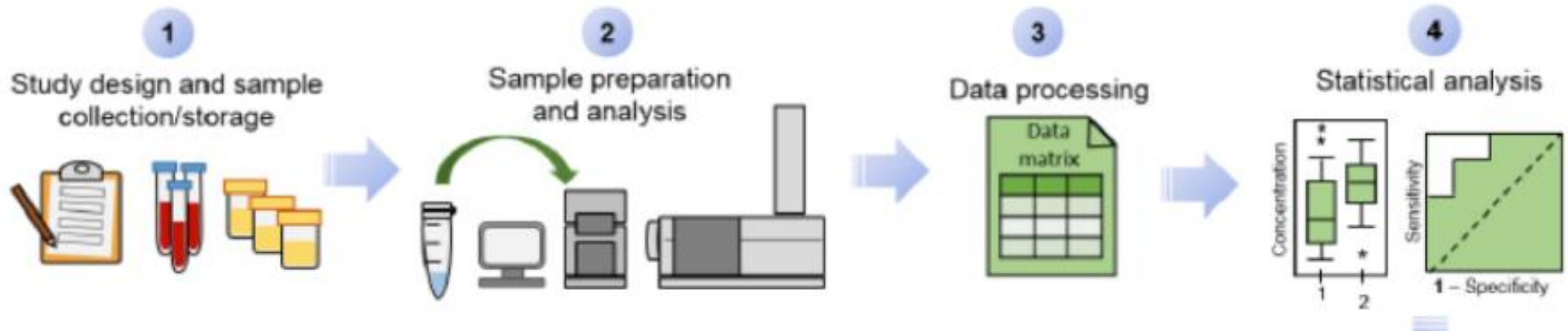


Xiaoqing Liu and Jason W. Locasale (2017). "Metabolomics: A Primer". Trends in Biochemical Sciences, Vol. 42, No. 4, pp. 274-284.

Metabolomics workflow



- Sample Extraction/Collection
- Separation
- Detection
- Data Extraction and analysis





Recap



- In this lecture, we have learned:
 - Basics of epigenomic mechanisms
 - ChIP-seq
 - ATAC-seq
 - Hi-C
 - Basics of proteomics techniques and related computational methods:
 - Basic mechanisms of mass spectrometry for protein identification
 - Metabolic profiling technologies continue to evolve
 - Mass Spectrometry Detection (MSD) is an active and evolving discipline
 - Metabolite identification remains challenging





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Appendix A: MACS Algorithm





- MACS is one of the most popular peak calling software. It was developed by Shirley Xiaole Liu's lab at Harvard University

Core idea of the algorithm

- The binding of TFs (transcription factors) to genome is a relatively random process, i.e. every position on the genome have the chance to be seen by a TF (however, with a different probability). Peak calling is aimed to find those hot spots that are easily seen
- It uses a Poisson distribution to assign p-values to peaks. But the distribution has a dynamic parameter λ , i.e. the event rate, to capture the influence of local biases

$$P(k \text{ events in interval}) = e^{-\lambda} \frac{\lambda^k}{k!}$$

- Here $\lambda = np$, $p = \frac{l}{s}$, n is the number of reads from sequencing, l is length of a single read, and s is length of the whole genome



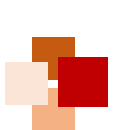


Sliding windows



- Given a sonication size (**bandwidth**) and a threshold of high-confidence fold-enrichment (**mfold**), MACS slides a window of length $2 \times \text{bandwidth}$ across the genome to find regions with tags more than mfold enriched relative to a random tag genome distribution
- Two parameters:
 - **mfold** specifies an interval of high-confidence enrichment ratio against the background on which to build the model. The default value in (10, 30) means that a model will be built on the basis of regions having read counts that are 10- to 30-fold of the background
 - **bandwidth** is half of the sliding window size used in the model-building step. It is set according to the length of the fragments expected experimentally from the sonication procedure





Estimate fragment size



Algorithm 1 Estimate Fragment Size

- 1: Slide a window of $2 \times \text{bandwidth}^3$ across genome
 - 2: Identify regions of moderate enrichment (mfold: 10-30 fold)
 - 3: **for each** peak i of 1000 randomly chosen enriched regions
 do
 - 4: separate reads into + and - strand
 - 5: Calculate mode of + and - summit
 - 6: $d_i \leftarrow |\text{mode}_+ - \text{mode}_-|$
 - 7: **end for**
 - 8: $d \leftarrow \text{average}_i(d_i)$
-

Thus, the distance between **bimodal** summits is assumed to be the estimated DNA fragment size d



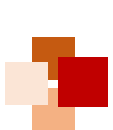


Strengths of MACS



- Most widely used peak caller
- Identifies genome-wide locations of TF binding, histone modification or NFRs (nucleosome-free regions) from ChIP-seq or ATAC-seq data
- Uses controls to eliminate biases due to GC content, mappability, DNA repeats or CNVs
- Can call narrow and broad peaks
- Many settings for optimizing results



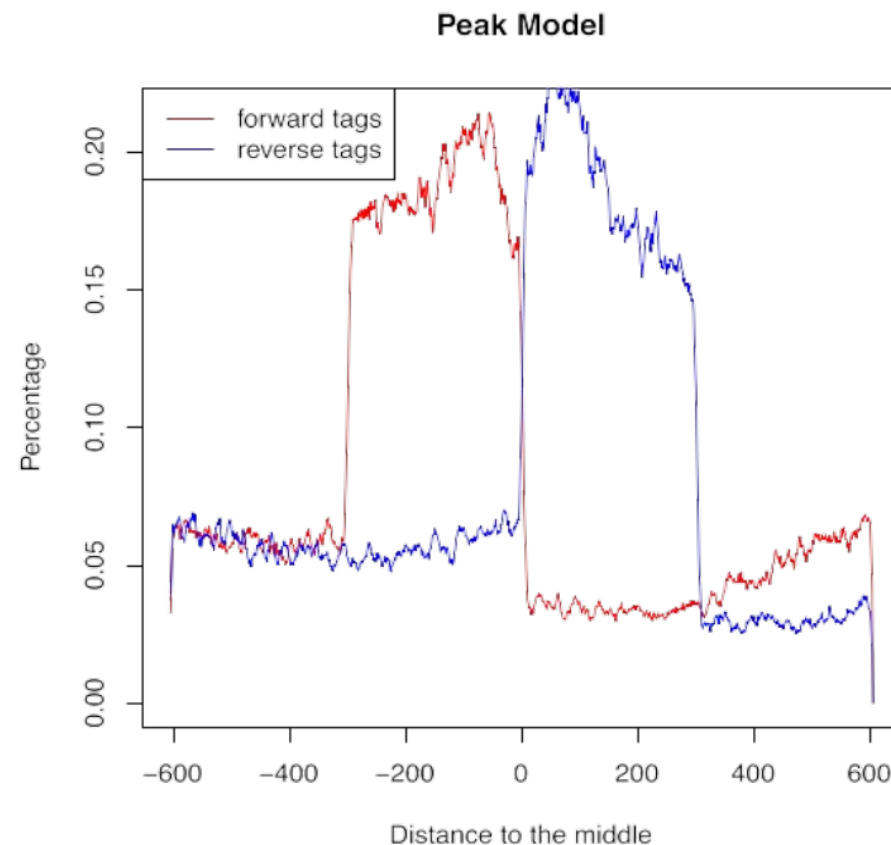


Peak calling with MACS (1)



Step 1: Estimate fragment length d and adjust read position

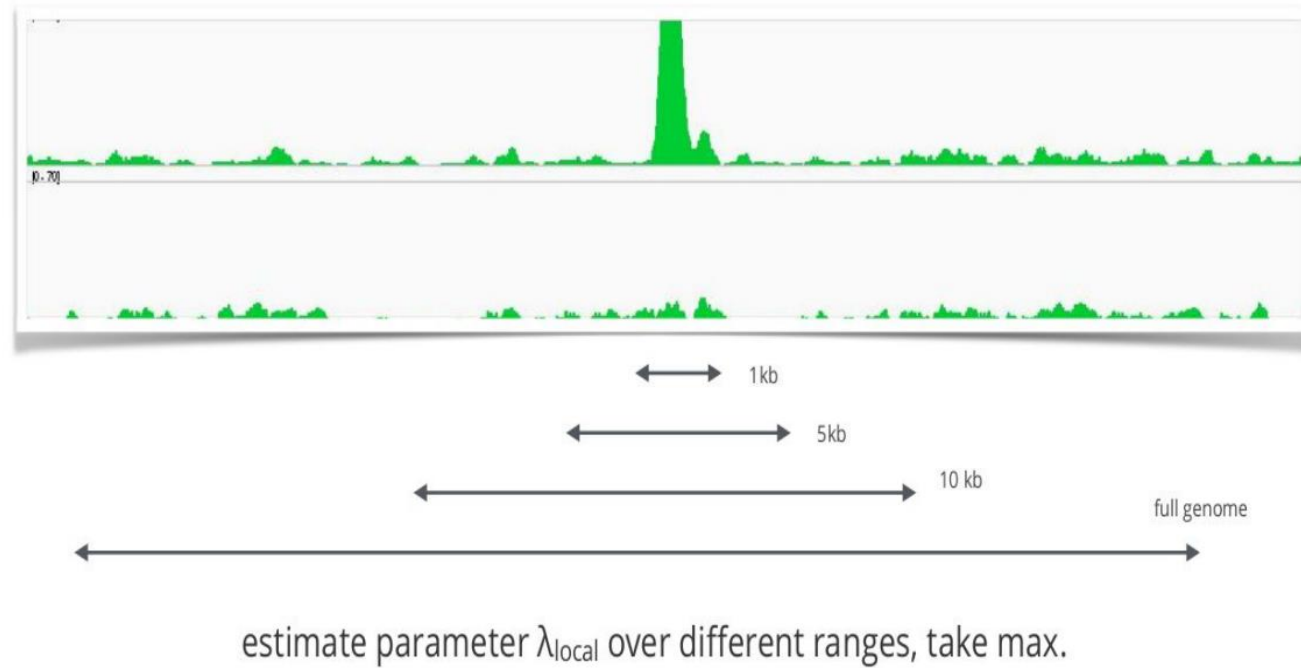
- Slide a window of length $2 * bw$ bandwidth (half of estimated sonication size) across genome
- Retain windows with (fold-enrichment of treatment / background) > mfold
- Compute the average +/- strand-specific read-densities for these bins

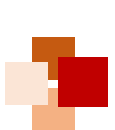


Peak calling with MACS (2)

Step 2: Identify local noise

- Slide a window of size $2 * d$ across treatment and input
- Estimate λ_{local} , a parameter of Poisson distribution



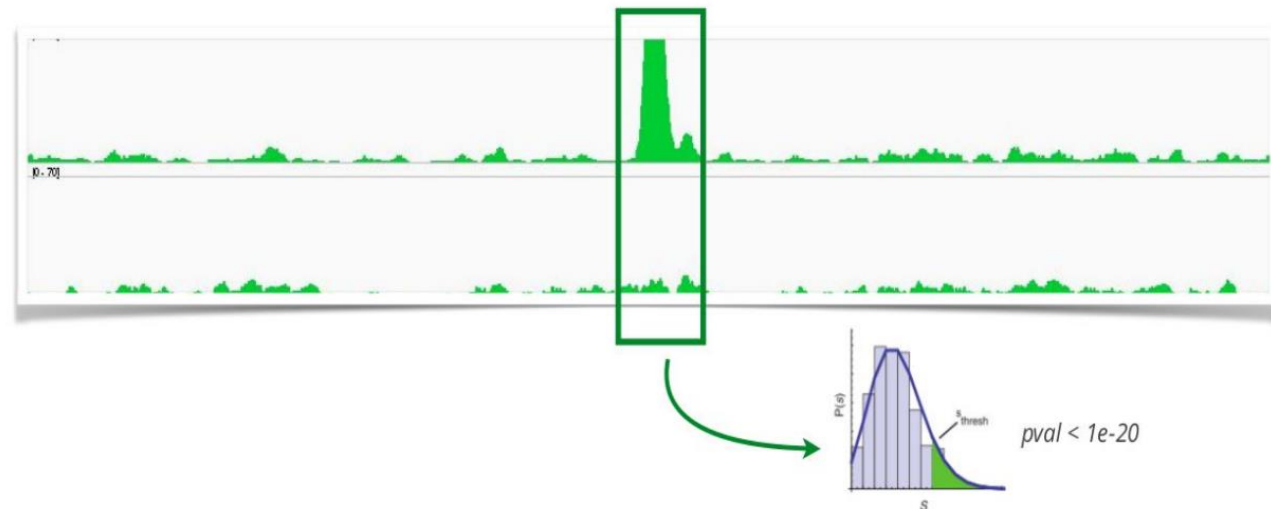


Peak calling with MACS (3)



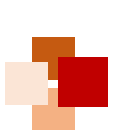
Step 3: Identify enriched (peak) regions

- Determine regions with p-value < a cut-off value (e.g. 10^{-20})
- Determine summit position within enriched regions as max density



estimate parameter λ_{local} over different ranges, take max.





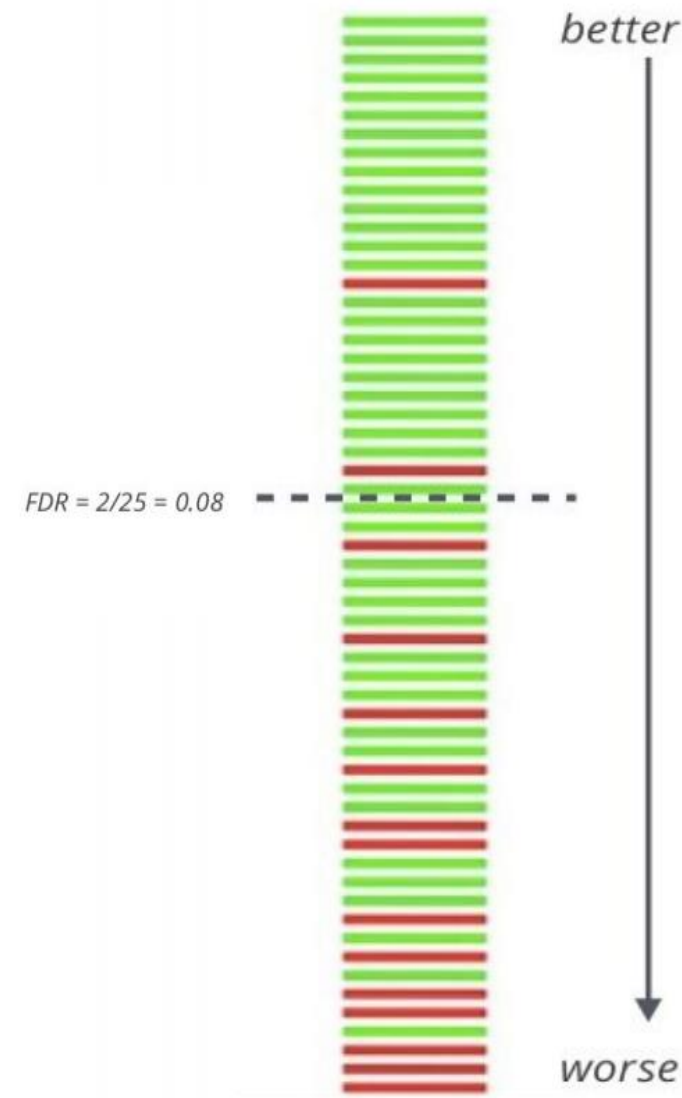
Peak calling with MACS (4)



Step 4: Estimate FDR

- Positive peaks (P-values)
- Swap treatment and input, call negative peaks (P-value)

$$\text{FDR} = \frac{\# \text{ negative peaks with pval} < p}{\# \text{ positive peaks with pval} < p}$$



■ From reads to peaks



- Positive and negative strand reads do not represent true binding sites
- Fragment length d can be detected experimentally or estimated from strand asymmetry in data
- Reads from both strands can be extended to the length of d
- Reads can be shifted towards 3' by $d/2$





Appendix B: De Novo Peptide Sequencing





De novo peptide sequencing problem

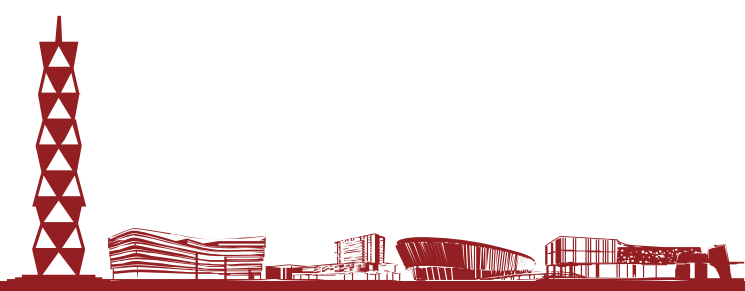


- **Input:**

- An MS/MS spectrum M
- The total mass wt of the peptide
- An error bound δ (default $\delta = 0.5$)

- **Output:**

- The peptide sequence





Assumption of the spectrum

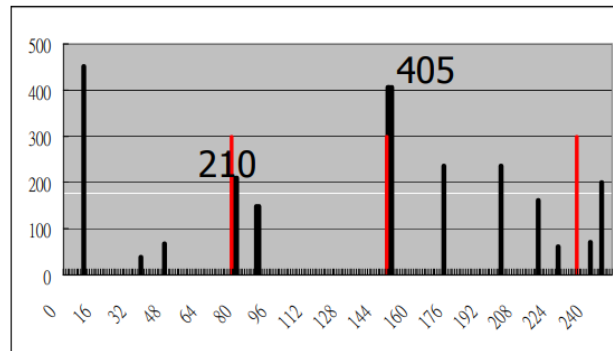


- We assume that all the ions are **singly** charged
- In fact, in an MS/MS experiment, an ion can be charged differently
- Fortunately, if a spectrum has peaks corresponding to multiply charged ions, there exists standard method to convert those peaks to their singly charged equivalents

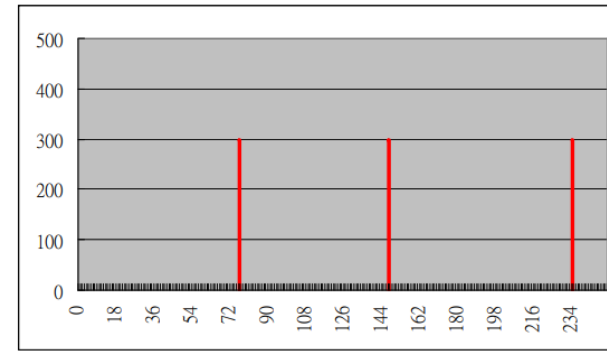


Simple scoring scheme

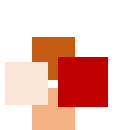
- Consider a peptide $P = a_1a_2...a_k$
 - Recall that y-ions are expected to have the highest intensities
 - If M is a spectrum for P , we can find peaks for $m/z = y_i$ for $i = 1, 2, \dots, k$
- So, we define the score function as:
$$\text{score}(M, P) = \sum \{h | (x, h) \in M, |x - y_i| \leq \delta \text{ for } i = 1, 2, \dots, k\}$$
- Example: $P = \text{SAG}$
 - $y_1 = 57.05 + 71.08 + 87.08 + 19 = 234.21$
 - $y_2 = 57.05 + 71.08 + 19 = 147.13$
 - $y_3 = 57.05 + 19 = 76.05$
- $\text{score}(M, P) = 210 + 405 = 615$



Black peaks: real peaks



Red peaks: artificial y-ions



Refined problem



- **Input:**

- An MS/MS spectrum M
- The total mass wt of the peptide
- An error bound δ

- **Output:**

- A peptide P such that $wt - \delta \leq w(P) \leq wt + \delta$, which maximizes $\text{score}(M, P)$





Brute-force solution



- For every possible peptide P such that $|w(P) - wt| \leq \delta$, compute $\text{score}(M, P)$
- Report the peptide P such that:
 - $|w(P) - wt| \leq \delta$, and
 - $\text{score}(M, P)$ is maximized
- The above algorithm takes **exponential time**. Very slow!
- Can we solve the problem faster? Yes, by dynamic programming (PEAKS algorithm in Appendix D)





Appendix C: Protein Database Search





Protein database searching problem



- **Input:**

- A database of protein (DB)
- A raw MS/MS spectrum (M)
- The mass wt of the peptide corresponding to M

- **Output:**

- A protein whose peptide is expected to have mass wt and a MS/MS spectrum similar to M

- In this lecture, let's look at a solution called SEQUEST (Eng et al. 1994)





- **Step 1:** Reduction of the tandem mass spectrometry data
 - To avoid noise, only 200 most abundant signals of the raw spectrum are used
 - Also, the total signals of the 200 signals are renormalized to 100
- **Step 2:** Search the protein database DB to find all peptides such that each peptide P has mass within $(wt \pm 1)$ Da





- **Step 3:** Rank the top 500 fit sequences by a specific scoring function
- **Step 4:** Compare the spectral similarity. Use cross-correlation analysis to generate the final score and rank the sequences
- The abundance of ions in the hypothetical spectrum:
 - 50 (b-ion, y-ion), 25 (mass/charge within ± 1 from b or y), or 10 (a-ion)



Other tasks of proteomics



- Protein Quantification
- Analysis of protein post-translational modifications (PTMs)
- Protein localization analysis
- Protein functional analysis
- Etc.





Appendix D: PEAKS algorithm





Idea of dynamic programming



- Try to identify the residues one by one from right to left
- Let $f_M(r) = \sum\{h | (x, h) \in M, |x - r| \leq \delta\}$
 - $f_M(r)$ is the sum of all peaks in M whose mass is close to r
- Observation:
 - $\text{score}(M, a_1 a_2 \dots a_k) =$
 $\text{score}(M, a_1 a_2 \dots a_{k-1}) + f_M(w(a_1 a_2 \dots a_k) + 19)$



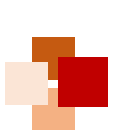


Simple dynamic programming solution



- Let $V(r)$ be the maximum score(M, P) among all possible peptide P such that $w(P) = r$
- Our aim is to find $\max_{|r-wt| \leq \delta} V(r)$. Then, by back-tracing, we can recover the peptide
- We have
 - $V(0) = 0$
 - $V(r) = \max_{a \in A} \{ V(r - w(a)) + f_M(r+19) \}$





Algorithm Max_Y_Ion

Require: The mass spectrum M and a weight W

Ensure: A peptide P of mass between $W - \delta$ and $W + \delta$ which maximizes $score_Y(M, P)$.

- 1: Set $V(r) = 0$ for $r < 0$
- 2: **for** $r = 0$ to $W + \delta$ **do**
- 3: $V(r) = \max_{a \in A} \{V(r - w(a)) + f_M(r + 19)\}$
- 4: **end for**
- 5: $r' = \arg \max_{W - \delta \leq r \leq W + \delta} V(r)$
- 6: Through back-tracing, we find the peptide P of mass r' which maximizes $score_Y(M, P)$



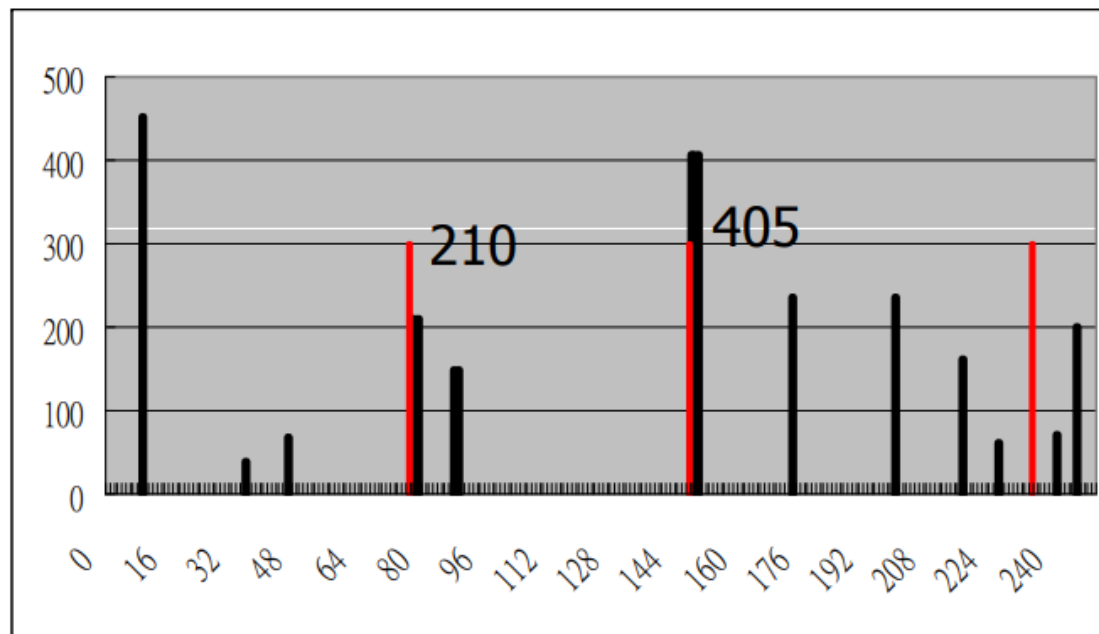


Example



- Given the spectrum M and $wt = 215.21$
 - $V(76.05) = V(0) + 210 = 210$ (due to G)
 - $V(147.13) = V(76.05) + 405 = 615$ (due to A)
 - $V(234.21) = V(147.13) + 0 = 615$ (due to S)
- By back-tracing, we recover SAG

M





Time analysis



- We need to fill in the V table with wt entries
- Each entry can be computed in $O(|A|)$ time
- So, the total time complexity is $O(|A|*wt)$ time



- Besides y-ions, we can also use information from b-ions
- Consider a peptide $P = a_1a_2...a_k$
 - If M is a spectrum for P , we can find peaks for $m/z = y_i$ or $m/z = b_i$ for $i = 1, 2, ..., k$
- So, we redefine the score function $\text{score}(M, P)$ as

$$\sum \{h | (x, h) \in M, |x - y_i| \leq \delta \text{ or } |x - b_i| \leq \delta \text{ for } i = 1, 2, \dots, k\}$$





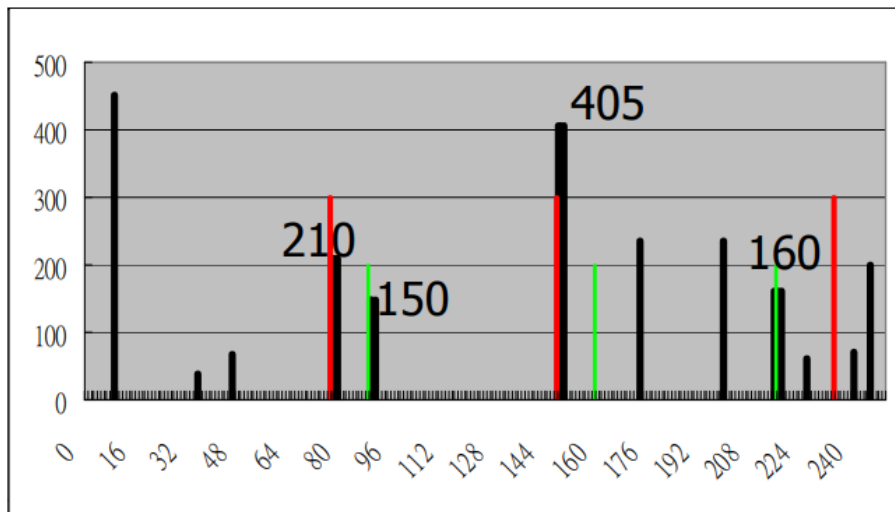
Better scoring scheme example



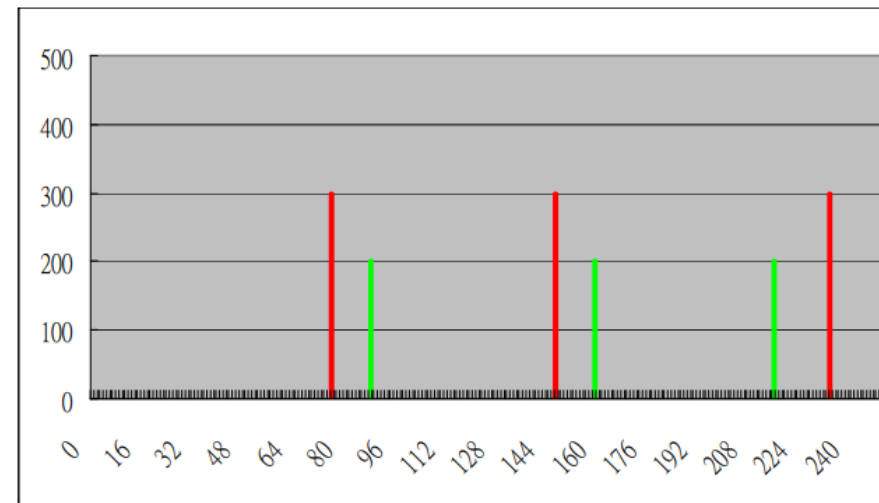
■ E.g. P=SAG

- $y_1 = 57.05 + 71.08 + 87.08 + 19 = 234.21$
- $y_2 = 57.05 + 71.08 + 19 = 147.13$
- $y_3 = 57.05 + 19 = 76.05$
- $b_1 = 87.08 + 1 = 88.08$
- $b_2 = 87.08 + 71.08 + 1 = 159.16$
- $b_3 = 87.08 + 71.08 + 57.05 + 1 = 216.21$

$$\begin{aligned}\text{Score}(M,P) \\ &= 210 + 405 + 150 + 160 \\ &= 925\end{aligned}$$



Black peaks: real peaks
Green peaks: artificial b-ions

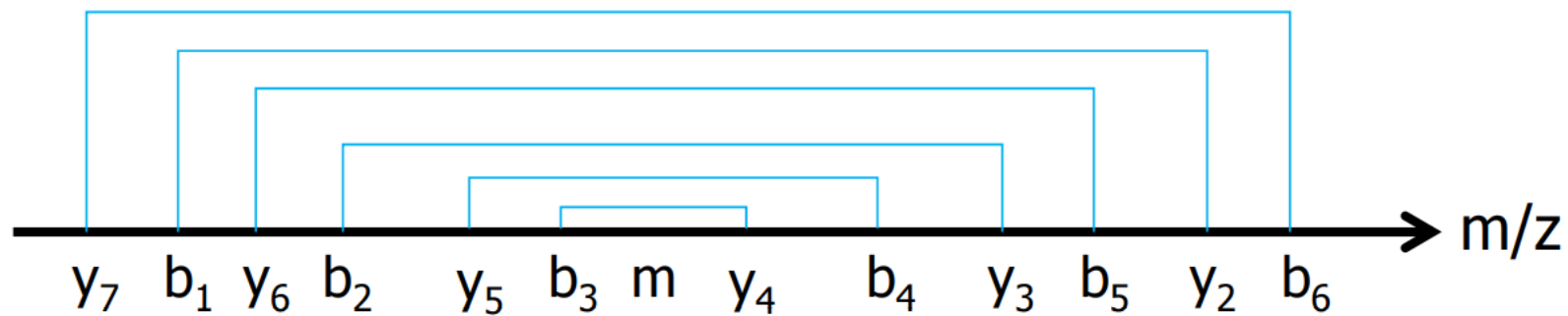


Red peaks: artificial y-ions



■ Observation (I)

- Suppose $P = a_1 a_2 \dots a_k$
- b_i is strictly increasing while y_j is strictly decreasing
 - Proof: For any peptide Q and amino acid a , $w(Qa), w(aQ) > w(Q)$
 - Hence, $b_{i+1} - b_i, y_j - y_{j+1} \geq \min_{a \in A} w(a) = 57.05 > 0$
- Note: $b_i + y_{i+1} = w(P) + 20$
 - Hence, (b_i, y_{i+1}) , for all $i=1, 2, \dots, k$, form a set of nested regions
 - For the adjacent nested intervals, the mass difference is at most $\max_{a \in A} w(a) = 186.21$

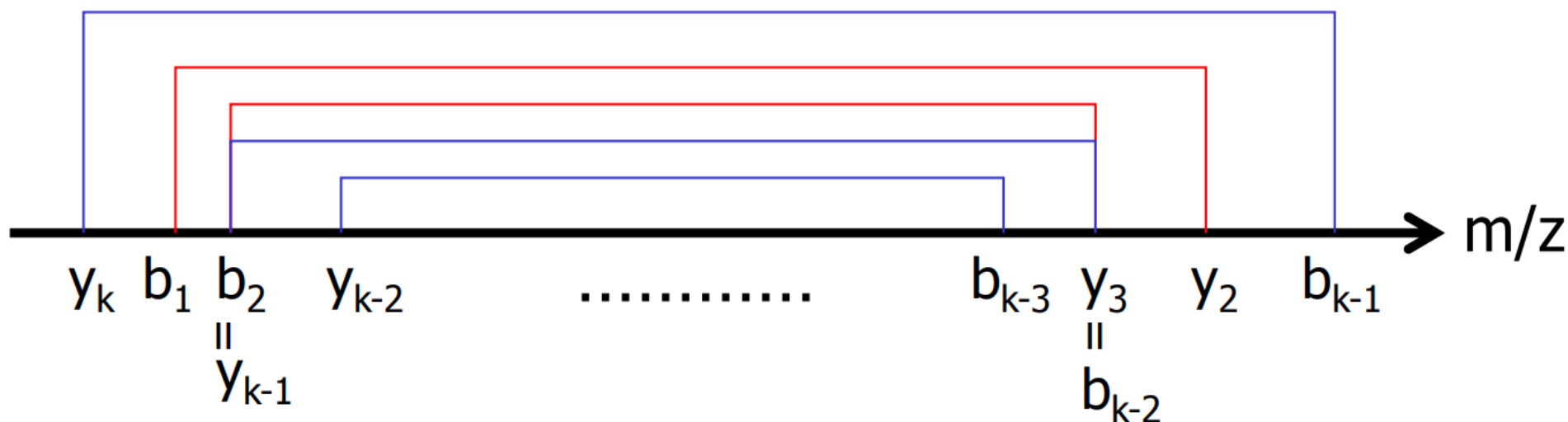


Consider $P = a_1 a_2 \dots a_7$.
 $m = (w(P) + 20) / 2$

Can we solve the problem using previous DP?



- No
- The reason is that, for some masses y_i and b_j , their masses may be very close and correspond to the same peak $(x, h) \in M$
- In this case, the previous DP will sum the same peaks twice





Observation (II)



- Note that the outer-most L intervals are formed by breaking the prefix $a_1 \dots a_i$ and the suffix $a_j \dots a_k$, where $i + (k - j + 1) = L$
- Let $\text{score}'(M, a_1 \dots a_i, a_j \dots a_k)$ be the sum of the intensities of all b-ion and y-ion peaks formed by breaking the peptide P between a_x and a_{x+1} for $x \in \{1, \dots, i\} \cup \{j-1, \dots, k-1\}$
- Let $f_M(r, s)$ be the sum of all peaks in M which are close to r and $\text{wt}+20-r$ but not close to s and $\text{wt}+20-s$ (this is used to avoid double counting)
- We have

$$\begin{aligned} & \text{score}'(M, a_1 \dots a_i, a_j \dots a_k) \\ &= \begin{cases} \text{score}'(M, a_1 \dots a_{i-1}, a_j \dots a_k) + f_M(b_i, y_j) & \text{if } b_i \geq y_j \\ \text{score}'(M, a_1 \dots a_i, a_{j+1} \dots a_k) + f_M(y_j, b_i) & \text{otherwise} \end{cases} \end{aligned}$$





Solution (a more complicated dynamic programming)

- Let $\hat{a}$ be $\max_{a \in A} w(a) = 186.21$
- For every $|r - s| \leq \hat{a}$, let $V(r, s)$ be the maximum score' (M, P_1, P_2) among all possible P_1 and P_2 , where $w(P_1) = r$ and $w(P_2) = s$

$$V(r, s) = \max \begin{cases} \max_{a \in A} \{V(r - w(a), s) + f_M(r + 1, s + 19)\}, & r \geq s \\ \max_{a \in A} \{V(r, s - w(a)) + f_M(s + 19, r + 1)\}, & r < s \end{cases}$$

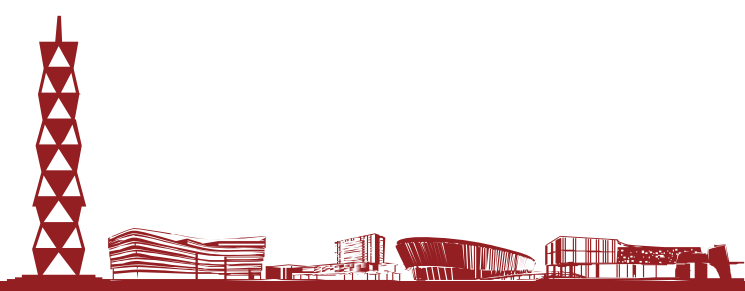
with base case $V(r, s) = 0 (r \leq 0, s \leq 0)$.





Solution (a more complicated dynamic programming)

- Aim: Find the best $V(r, s)$ such that $wt + 20 = r + s + w(a)$ for some $a \in A$
- Then, by back-tracking, we can recover the peptide



Algorithm Sandwich

Require: A mass spectrum M , a weight W , and an error bound δ

Ensure: A peptide P such that $score(M, P)$ is maximized and $|w(P) - W| \leq \delta$

- 1: Let $\hat{a} = \max_{a \in A} w(a)$
- 2: Initialize all $V(r, s) = -\infty$; Let $V(0, 0) = 0$
- 3: **for** $r = 1$ to $(W/2 + \hat{a})$ **do**
- 4: **for** $s = r - \hat{a}$ to $\min\{r + \hat{a}, W - r\}$ **do**
- 5: **for** $a \in A$ such that $r + s + w(a) < W$ **do**
- 6: **if** $r < s$ **then**
- 7: $V(r, s) = \max\{V(r, s), V(r - w(a), s) + f_M(r + 1, s + 19)\}$
- 8: **else**
- 9: $V(r, s) = \max\{V(r, s), V(r, s - w(a)) + f_M(s + 19, r + 1)\}$
- 10: **end if**
- 11: **end for**
- 12: **end for**
- 13: **end for**
- 14: Identify the best $V(r, s)$ among all r, s, a satisfying $|r - s| < \hat{a}$ and $|r + s + w(a) - W| < \delta$. Through back-tracing, we can recover a peptide $P = P' a P''$ where $w(P') = r$ and $w(P'') = s$.



Spectrum graph approach



- Another method to recover the peptide is based on spectrum graph, which is defined as follows
- For each mass r in the spectrum M , we generate two vertices of masses r and $wt - r$
- We also include 2 additional vertices
 - Starting vertex with mass = 0, and
 - Ending vertex with mass = wt





Generating edges in the spectrum graph



- For every pair of masses r and s , if $r - s$ equals the mass of an amino acid A , we connect x and y with an edge of label A
- Since there may be some missing peaks in the spectrum
 - If $r - s$ equals the total mass of two amino acids A_1A_2 , we connect x and y with an edge of label A_1A_2
 - If $r - s$ equals the total mass of three amino acids $A_1A_2A_3$, we connect x and y with an edge of label $A_1A_2A_3$



Meaning of a path in the graph

- Every path from start to end corresponds to a possible peptide in the spectrum
- However, there could be many possible paths
- Observe that a vertex has higher probability to be real if all ion types are available
- Hence, we can assign a score depending on whether some ion types are missing
- Then, this is a problem of finding the heaviest path, which can be solved in polynomial time

